

Social Experience Shapes Fighting Strategies for Reproductive Success

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The paper presents a new behavioral assay for *Drosophila* aggression and demonstrates that social experience influences fighting strategies, with group-housed males favoring high-intensity but low-frequency tussling over aggressive lunging observed in isolated males. This paper is **important** for researchers studying the impact of social isolation on aggression, while the description of tussling behavior and the interpretation of the link between tussling and mating success are **incomplete**.

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Abstract

Social isolation generally increases aggression but decreases mating competition, resulting in an intricate and ambiguous relationship between social experience, aggression, and reproductive success. In male *Drosophila*, aggression is often characterized by lunging, a frequent and comparatively gentle combat behavior. Here, we provide a behavioral paradigm for studying a less frequent but more vigorous fighting form known as tussling. While social enrichment decreases lunging, aligning with past observations, it heightens the more forceful tussling behavior. These two forms of aggression rely on different olfactory receptor neurons, specifically Or67d for lunging and Or47b for tussling. We further identify three pairs of central pC1 neurons that specifically promote tussling. Moreover, shifting from lunging to tussling in socially enriched males is accompanied with better territory control and mating success. Our findings identify distinct sensory and central neurons for two fighting forms and suggest how social experience shapes fighting strategies to optimize reproductive success.

Introduction

Competition for resources such as territory, food, and mates is crucial for animal survival and reproduction (Bergman et al., 2010 [↗](#); Hoffmann and Cacoiani, 1990 [↗](#); Stockley and Campbell, 2013 [↗](#); Zwarts et al., 2012 [↗](#)). Among diverse competitive strategies, physical combat is a

prevalent form of fighting to repel rivals and regulated by both genetic factors and social environment (Chen, 2022 [↗](#); Zwarts et al., 2012 [↗](#)). Understanding how social experience influences combat behaviors in animals to maximize their survival and reproduction remains a fundamentally important yet inadequately understood question.

In the fruit fly, *Drosophila melanogaster*, inter-male aggression includes relative low-intensity fights such as lunging, in which one fly rears up on its hind legs and snaps down onto its opponent, as well as relative high-intensity fights like tussling, where two males tumble over each other (Certel and Kravitz, 2012 [↗](#); Chen et al., 2002 [↗](#); Hoopfer, 2016 [↗](#); Palavicino-Maggio and Sengupta, 2022 [↗](#)). A recent work showed that lunges are insufficient to establish dominance but mainly serves to maintain social status, while tussling frequency is correlated with dominance establishment (Simon and Heberlein, 2020 [↗](#)). It is noteworthy that the majority of studies focus on the low-intensity lunging behavior, owing to its higher frequency and ease of observation. Substantial studies have identified regulatory genes and the neural circuit underlying the lunging-based aggressive behaviors (Hoopfer, 2016 [↗](#); Kravitz and Fernandez, 2015 [↗](#)). In particular, the male-specific volatile pheromone 11-*cis*-vaccenyl acetate (cVA) acutely promotes lunging behavior via its olfactory receptor Or67d (Wang and Anderson, 2010 [↗](#)). Interestingly, chronic cVA exposure in socially-enriched males reduces the level of lunging via another olfactory receptor Or65a (Liu et al., 2011 [↗](#)). Consistently, it has been found that social isolation increases lunging frequency (Palavicino-Maggio and Sengupta, 2022 [↗](#); Yadav et al., 2024 [↗](#)) by altering the expression of genes such as *Cyp6a20* (Wang et al., 2008 [↗](#)), *Drosulfakinin* (*Dsk*) (Agrawal et al., 2020 [↗](#)), and *Tachykinin* (*Tk*) (Asahina et al., 2014 [↗](#)). Social isolation also increases male aggression in mice (Matsumoto et al., 2005 [↗](#); Zelikowsky et al., 2018 [↗](#)) and even in humans (Ferguson et al., 2005 [↗](#)). In mice, the TK homologue Tachykinin 2 (*Tac2*) responds to social isolation and promotes inter-male aggressive behaviors (Zelikowsky et al., 2018 [↗](#)). These findings contribute to the consensus that social isolation escalates aggression, whereas social enrichment mitigates it in animal models.

Meanwhile, social enrichment increases male mating success by enhancing the sensitivity of *Or47b*-expressing olfactory receptor neurons (ORNs) to fly pheromones common to both sexes (Lin et al., 2016 [↗](#); Sethi et al., 2019 [↗](#)). Artificially activating *Or47b* ORNs also improves mating advantage in a competitive assay (Dweck et al., 2015 [↗](#)). Group housing and age-dependent juvenile hormone (JH) jointly increase the expression of the male-specific proteins from *fruitless* (*fru^M*), which is a master gene for male courtship behaviors (Peng et al., 2021 [↗](#); Sato et al., 2019 [↗](#); Yamamoto and Koganezawa, 2013 [↗](#)), in the *Or47b*-expressing ORNs (Hueston et al., 2016 [↗](#); Sethi et al., 2019 [↗](#); Zhao et al., 2020 [↗](#)). Notably, courtship levels in single-housed (SH) wild-type males are equal to (Dankert et al., 2009 [↗](#); Pan and Baker, 2014 [↗](#)), in some cases, even stronger than those in group-housed (GH) males (Dankert et al., 2009 [↗](#)). In fact, group-housing decreases excitability of the courtship-promoting P1 neurons (Inagaki et al., 2014 [↗](#)), which express both terminal genes in the sex determination hierarchy, *fru* and *doublesex* (*dsx*). These findings raise a question on how social enrichment, which suppresses lunging-based aggression and does not increase courtship intensity, would facilitate mating competition.

Here, we modify a behavioral paradigm for studying inter-male tussling in *Drosophila* and discover that this infrequent yet intense form of combat is enhanced by social enrichment and is crucial for territory dominance and mating competition. This process requires olfactory receptor Or47b, but not Or67d or Or65a. We further identify three pairs of central *dsx*-expressing pC1 neurons that specifically promote tussling. Moreover, we demonstrate that socially enriched males can overcome mating disadvantages associated with aging.

Results

A Paradigm for High-intensity Aggression in *Drosophila*

While studying the lunging-based aggression in *Drosophila* using a high-throughput aggression assay based on previously aggression paradigms (Certeel and Kravitz, 2012 [↗](#); Wu et al., 2020 [↗](#); Zhou et al., 2008 [↗](#)), we occasionally observed the high-intensity boxing and tussling behavior in male flies (Chen et al., 2002 [↗](#); Nilsen et al., 2004 [↗](#)), which are difficult to distinguish and hereafter simply referred to as tussling. We reasoned that the intensity, instead of the frequency, could be decisive for a combat (Simon and Heberlein, 2020 [↗](#)). Thus, although tussling is a less frequent behavior, it could serve a vital function.

To induce a higher level of tussling for further manipulation, we first introduced a fixed live virgin female in the same behavioral chamber for assaying lunging. The female fly was immobilized in the middle of the food, with its head embedded in the food and its body exposed to enhance male competition (Figure 1A [↗](#)). We found that males that were group-housed for 7 days showed a significantly higher frequency of tussling behavior and reduced latency to tussle in the presence of an embedded virgin female (Figure 1B [↗](#)). We next compared tussling in male flies of different ages. We observed a significantly higher proportion of 14-day-old males raised in groups that showed tussling behavior during the 2-hour observation period, compared to the 7-day-old males (Figure 1B [↗](#)). Furthermore, 2-day-old males never displayed tussling behavior. In contrast, 14-day-old males showed low level of lunging behavior than 7-day-old males did (Figure 1-figure supplement 1 [↗](#)). Previous studies have shown that, in addition to mating resources, food quality is another crucial factor affecting aggressive behaviors (Lim et al., 2014 [↗](#)). Therefore, we next assayed tussling behavior in 14-day-old males in the presence of an embedded virgin female with different concentrations of sucrose or yeast in the food during the behavioral test. We found that these males displayed similar levels of tussling behavior with different concentrations of sucrose in the food (Figure 1C [↗](#)). On the contrary, male tussling intensity increased significantly with higher yeast concentrations such that more than 70% of males displayed this high-intensity fighting behavior (Figures 1D [↗](#), see Figure 1-video supplement 1). Taken together, these results provide an efficient behavioral paradigm for the high-intensity tussling behavior and indicate that older males display more tussling but less lunging than younger ones.

Social Enrichment Inhibits Low-intensity Lunging but Enhances High-intensity Tussling

Previous studies have shown that social enrichment inhibits aggressive behaviors, specifically the low-intensity lunging, in male flies (Yadav et al., 2024 [↗](#)). Therefore, we wondered whether social enrichment might also inhibit the high-intensity tussling behavior in males. To answer this question, we compared both fighting behaviors in SH and GH 14-day-old males (Figure 2A [↗](#)). Lunging and tussling events from each group of representative samples were exhibited in raster plots for comparison (Figure 2B [↗](#)). We scored lunging behavior mainly in the first 10-min of test, due to its relatively higher occurrence, and tussling behavior during the whole 2-hour observational period since males take much longer time to initiate tussling (Figure 1D [↗](#)). Interestingly, we found that while SH males performed more frequent lunging behavior in 10 min than GH males did, which is consistent with previous findings, they rarely showed tussling behavior. In contrast, GH males displayed a large number of tussling events within 2 hours (Figure 2B [↗](#)). Statistical analysis of tussling and lunging behaviors showed that GH males exhibited reduced tussling latency and enhanced tussling frequency, in addition to increased lunging latency and decreased lunging frequency, compared to SH males (Figures 2C-F [↗](#)). We also quantified the duration of each tussling and lunging events, and found that the average duration of individual lunging events was less than 0.2 second, while a tussling event lasted from seconds to minutes (Figure 2G [↗](#)). This further suggests that tussling is a more intense form of aggressive behavior compared to lunging. These results indicate that social enrichment inhibits low-intensity lunging but promotes high-intensity tussling in male flies.

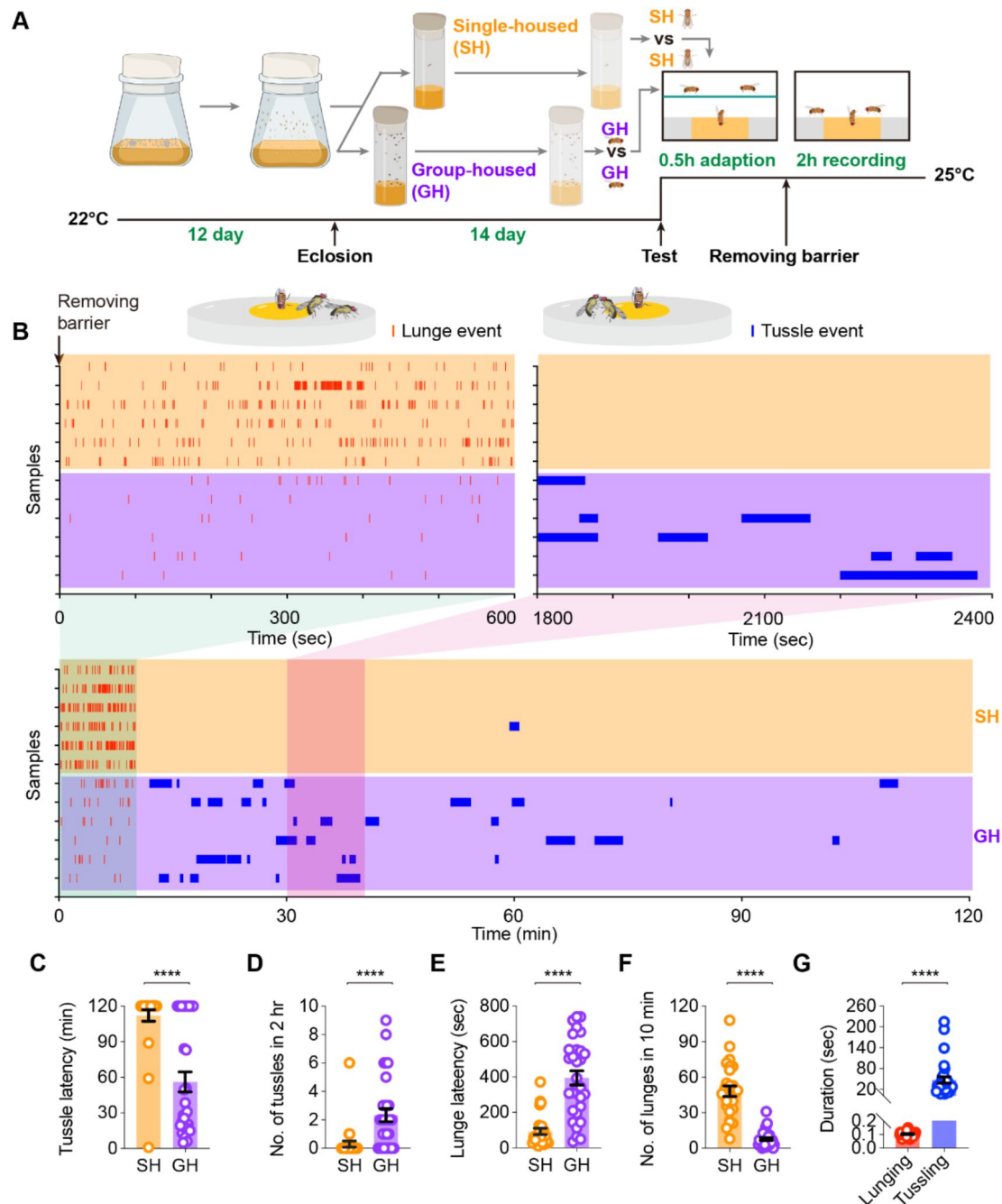


Figure 2.

Social enrichment inhibits lunging behavior but promotes tussling behavior in males.

(A) Schematic diagram of *Drosophila* rearing conditions and experimental procedure.

(B) Representative raster plots illustrating lunge events in the first 10 minutes and tussle events over 2 hours in single-housed (SH) and group-housed (GH) males.

(C, D) Latency (C) and number (D) of tussling behavior in SH and GH males, n=27, 30 for SH and GH males, respectively, ****P < 0.0001, Mann-Whitney test for both.

(E, F) Latency (E) and number (F) of lunging behavior in SH and GH males, n=27, 30 for SH and GH males, respectively, ****P < 0.0001, Mann-Whitney test for both. Error bars correspond to SEM.

(G) Average duration of lunging events in the first 10 minutes and tussling events within 2 hours in GH males, n=30 events from 10 samples for each group, ****P < 0.0001, Mann-Whitney test. Error bars correspond to SEM.

Or47b ORNs Mediate Social Experience-induced Tussling Behavior

To further investigate the sensory basis underlying the opposite regulation of lunging and tussling behaviors by social experience, we tried to examine the role of cVA-sensitive neurons, as well as other *fru^M*-expressing ORNs, in mediating intermale aggression. Two cVA sensory neurons, *Or67d* and *Or65a* ORNs, play crucial roles in enhancing the lunging-based aggression in SH males, in which *Or67d* ORNs mediate acute cVA stimulation and promote male aggression (Wang and Anderson, 2010 [↗](#)), while *Or65a* ORNs mediate chronic cVA stimulation and inhibit male aggression in GH males (Liu et al., 2011 [↗](#)). In addition, the *fru^M*-expressing *Or47b* ORNs are sensitive to a class of pheromones common to both sexes, such as palmitoleic acid (PA) (Lin et al., 2016 [↗](#)), and their sensitivity is enhanced under group rearing conditions (Sethi et al., 2019 [↗](#)). *Ir84a* ORNs, which are also *fru^M*-positive, are involved in the perception of mixed food odors and promote male courtship behavior (Grosjean et al., 2011 [↗](#)). We silenced *Or67d*, *Or65a*, *Or47b* or *Ir84a* ORNs by expressing the inwardly rectifying potassium channel Kir2.1 (Baines et al., 2001 [↗](#)) and assayed lunging and tussling behaviors in 14-day-old GH males. We found that silencing *Or47b* ORNs, but not *Or67d*, *Or65a* or *Ir84a* ORNs, significantly decreased male tussling, compared to the control (Figure 3A, B [↗](#)). In contrast, silencing *Or67d* ORNs, but not other ORNs, significantly reduced male lunging (Figure 3C, D [↗](#)). These loss-of-function results suggest that *Or47b* ORNs mediate male tussling behavior, whereas *Or67d* ORNs regulate male lunging behavior.

To elucidate the potential role of the *Or47b* receptor in mediating tussling behavior, we knocked down its expression in *Or47b* ORNs using RNA interference (RNAi). Our results showed a significant decrease in male tussling behavior when the *Or47b* receptor expression was reduced (Figure 3E, F [↗](#)), indicating that the functionality of *Or47b* ORNs in tussling is dependent on the *Or47b* receptor. As previous studies suggested that the male-specific *fru^M* sustains *Or47b* expression in an activity-dependent manner (Sethi et al., 2019 [↗](#)), we next expressed microRNAs targeting *fru^M* (*UAS-fruMi*) (Meissner et al., 2016 [↗](#)) in *Or47b* ORNs and assayed male tussling. The efficiency of *Or47b* and *fru^M* RNAi knockdown was both validated by quantitative PCR (Figure 3-figure supplement 1 [↗](#)). We observed a slight increase of tussling latency in *fru^M* knocked-down males, although not significantly different from the control (Figure 3E, F [↗](#)), suggesting that *Or47b*, rather than *Fru^M*, plays a more crucial role in mediating male tussling.

To further confirm that the effect of social enrichment on tussling depends on the activity of *Or47b* ORNs, we artificially activated *Or47b* ORNs by expressing the bacterially derived sodium channel (NaChBac) (Ren et al., 2001 [↗](#)) in 14-day-old SH males. We found that these SH males with enhanced activity of *Or47b* ORNs showed a significant increase in tussling behavior compared to control SH males. Interestingly, the level of tussling in these SH males with activated *Or47b* ORNs was indistinguishable from that in GH males (Figure 3G, H [↗](#)). These gain-of-function results indicate that social enrichment acts through *Or47b* ORNs to promote male tussling. Using the trans-Tango technique that labels downstream neurons in a usage-dependent manner (Talay et al., 2017 [↗](#)), we found substantially more downstream signals of *Or47b* ORNs in GH males than those in SH males (Figure 3-figure supplement 2 [↗](#)), further supporting the involvement of *Or47b* ORNs under social enrichment.

Distinct Central Brain Neurons for Lunging and Tussling

We next set out to identify the central neurons involved in regulating the high-intensity tussling. We found that silencing the mushroom body neurons by expressing an inwardly rectifying potassium channel Kir2.1 (Baines et al., 2001 [↗](#)) driven by *R19B03-GAL4* did not affect male tussling, suggesting that the experience-dependent tussling behavior is not dependent on the classical learning and memory center. Surprisingly, inactivation of the previously-identified aggression-promoting P1^a and TK neurons did not affect tussling behavior (Figure 4A, B [↗](#)). These results imply that the central circuit for tussling behavior may be distinct from the previously

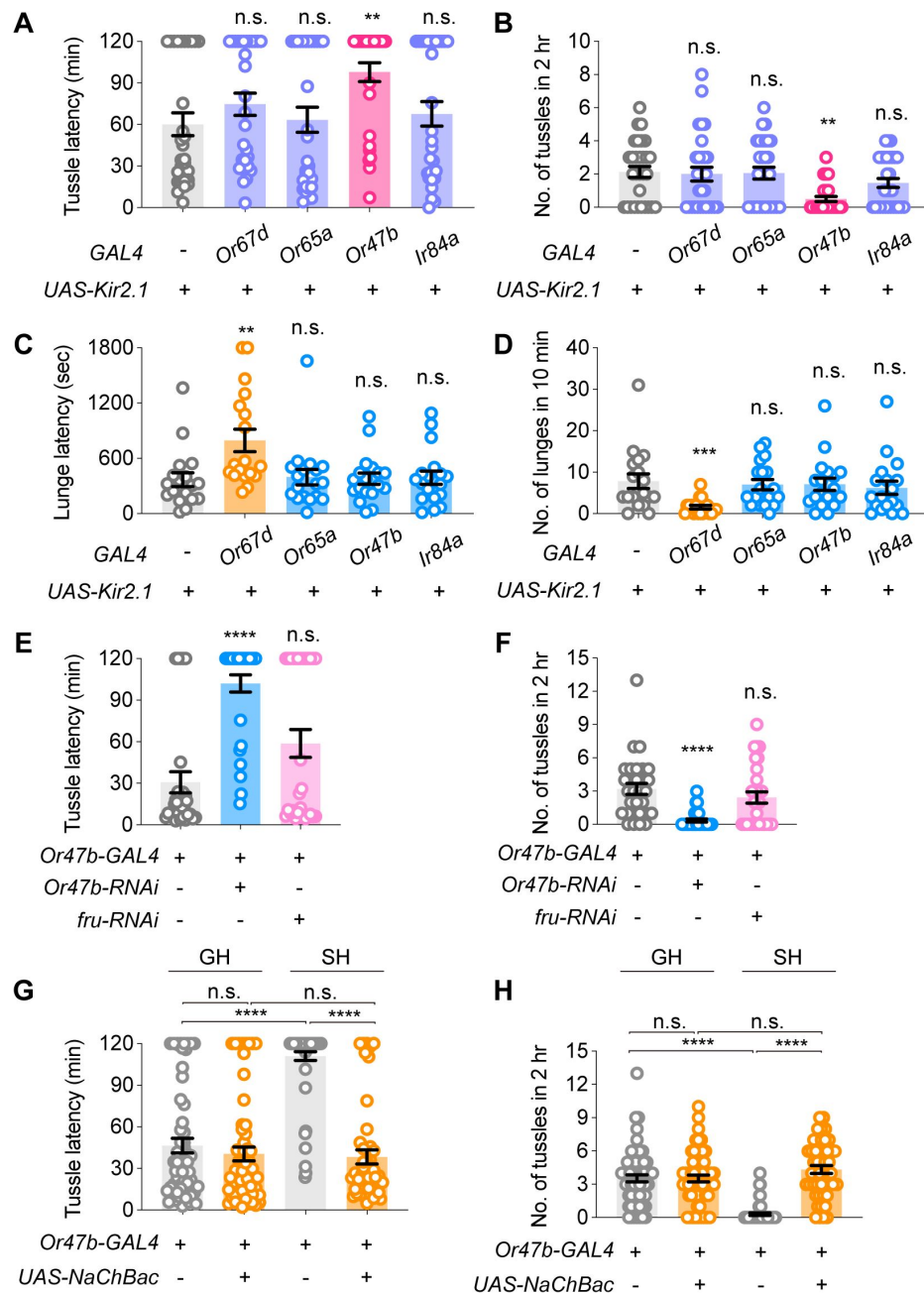


Figure 3.

Or47b neurons are required for male tussling but not lunging.

(A, B) The effect of inactivating different sensory neurons on latency (A) and number (B) of tussling behavior in 14-day-old group-housed males (G14). $n=30$ for each group, n.s., not significant, $**P < 0.01$, Kruskal-Wallis test.

(C, D) The effect of inactivating different sensory neurons on the latency (C) and number (D) of lunging behavior in G14 males. $n=30$ for each group, n.s., not significant, $**P < 0.01$, $***P < 0.001$, Kruskal-Wallis test.

(E, F) The effect of knocking down *Or47b* or *fru*^M expression in *Or47b* ORNs on the latency (E) and number (F) of tussling behavior in G14 males, respectively. $n=30$ for each group, n.s., not significant, $****P < 0.0001$, Kruskal-Wallis test.

(G, H) The effect of activating *Or47b* ORNs on latency (G) and number (H) of tussling behavior under conditions of GH or SH for 14 days, $n=60, 60, 60, 49$ respectively, n.s., not significant, $****P < 0.0001$, Kruskal-Wallis test. Error bars correspond to SEM.

reported aggression circuits. As *dsx*-expressing pC1 neurons, also termed P1 if co-expressing *fru*^M, are crucial for sexual and aggressive behaviors in both males and females, we also silenced two subsets of pC1 neurons (pC1^{SS1} and pC1^{SS2}), which were sexually dimorphic, and their functions were previously characterized in females but not in males (Wang et al., 2021 [4](#)). We found that inactivation of pC1^{SS2} but not pC1^{SS1} neurons significantly reduced the tussling behavior in male flies (**Figure 4A, B** [4](#)). The cell bodies of pC1^{SS2} and P1^a neurons are both located in the center of posterior brain and project to overlapping but distinct parts of the lateral protocerebral complex (**Figure 4C-E** [4](#)). These results suggest that pC1^{SS2} neurons may be key central neurons regulating tussling behavior.

To further compare the function of P1^a and pC1^{SS2} neurons in regulating male behaviors, we performed optogenetic activation experiments on these neurons expressing *CsChrimson*. Compared to control male pairs that did not show any significant change upon red light stimulation (**Figure 4F-H** [4](#)), optogenetic activation of P1^a neurons induced acute courtship behavior as indicated by unilateral wing extension (UWE), and upon red light removal, there was a significant increase in lunging but not tussling (**Figure 4I-K** [4](#)), consistent with previous findings. In contrast, optogenetic activation of pC1^{SS2} neurons induced strong and acute tussling behavior, with more than 50% of the time during red light in tussling, and the tussling behavior ceased immediately upon removal of red light (**Figure 4L** [4](#), see Figure 4-video supplement 1). Activation of pC1^{SS2} neurons did not significantly affect either male lunging or courtship (**Figure 4M, N** [4](#)). These results indicate that pC1^{SS2} neurons acutely promote the high-intensity tussling behavior.

Previous studies have revealed a crucial role of P1^a neurons in promoting a persistent internal state that enhances male aggression (Hoopfer et al., 2015 [4](#)). To test whether P1^a neurons can promote both lunging and tussling behaviors, we further performed thermal activation experiments. We found that activation of P1^a neurons via dTrpA1 at 30°C induced male tussling, just like activation of pC1^{SS2} neurons did (**Figure 4-figure supplement 1** [4](#)). As thermal activation but not optogenetic activation of P1^a neurons induced tussling, we reasoned that vision might be a key factor for this discrepancy, since all optogenetic experiments were performed in dark or red light. Thus, we re-performed these thermal activation experiments in dark and found that activation of pC1^{SS2} but not P1^a neurons induced tussling, although the level of tussling by pC1^{SS2}-activated males was lower compared to that in light, suggesting a crucial role of vision in male tussling. These optogenetic and thermogenetic activation experiments suggest that P1^a neurons can promote both lunging and tussling behaviors, but their function on tussling requires visual stimulation, while pC1^{SS2} neurons play a more direct role in acutely promoting male tussling.

We next used *retro*-Tango (Sorkaç et al., 2023 [4](#)) and *trans*-Tango (Talay et al., 2017 [4](#)) techniques to explore the potential upstream and downstream circuits of P1^a and pC1^{SS2} neurons, and observed substantial differences in both conditions (**Figure 4-figure supplement 2** [4](#)), suggesting largely parallel inputs and outputs for P1^a and pC1^{SS2} neurons.

Dsx^M Regulates the Development and Function of pC1^{SS2} Neurons

To further compare the differences between P1^a neurons and pC1^{SS2} neurons, we investigated whether P1^a and pC1^{SS2} neurons express *fru* and/or *dsx* using Fru^M and Dsx^M antibodies (Chen et al., 2021 [4](#); Peng et al., 2022 [4](#)). We found that most P1^a neurons were *fru*-positive and *dsx*-positive, while pC1^{SS2} neurons were *dsx*-positive but *fru*-negative (**Figure 5A** [4](#)-**5D** [4](#)). These results suggest that *dsx* instead of *fru* may play crucial roles in the development and function of pC1^{SS2} neurons.

To investigate the function of *dsx* in pC1^{SS2} neurons, we knocked down *dsx* via RNAi and compared the morphological and functional changes of these neurons. We found that *dsx*^M knockdown in pC1^{SS2} neurons slightly increased the number of cells from 2-3 pairs to 3-4 pairs

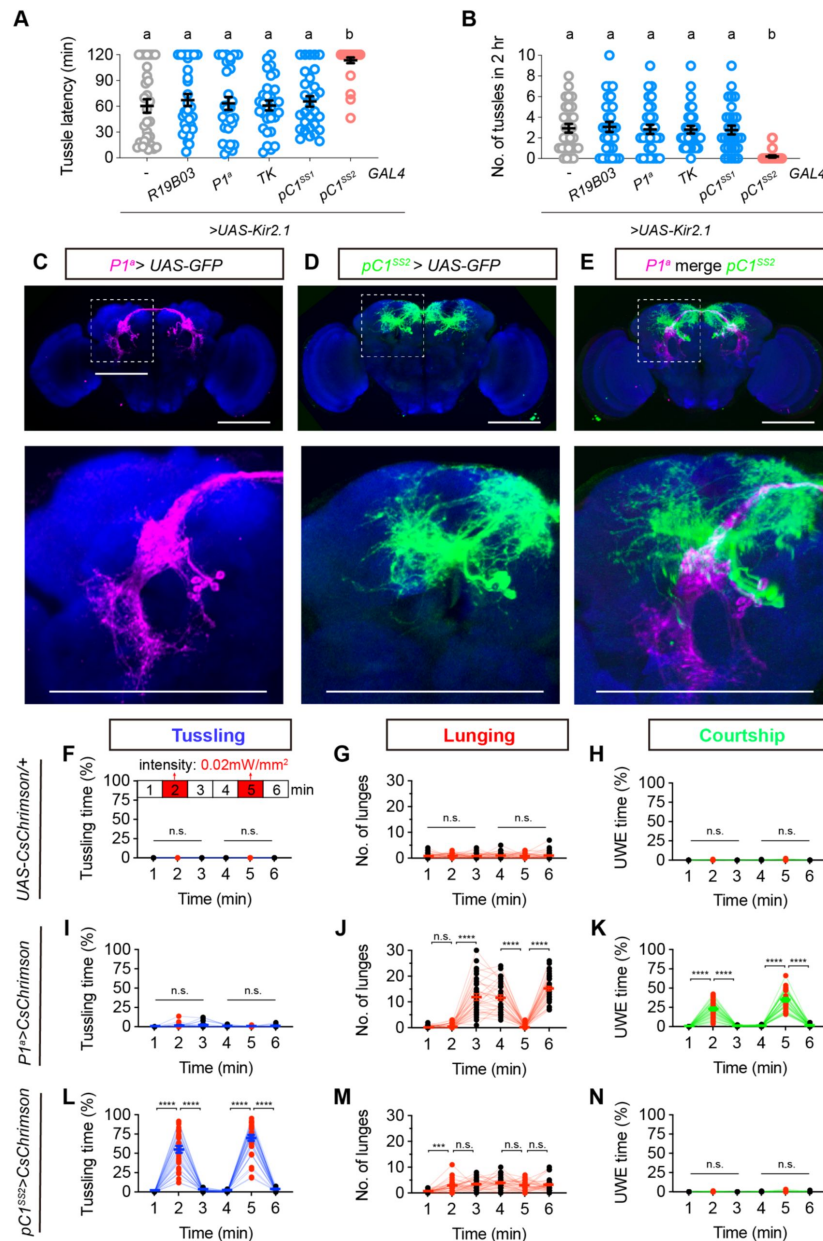


Figure 4.

A subset of pC1 neurons specifically promote male tussling behavior.

(A, B) The effect of inactivating candidate interneurons on latency (A) and number (B) of male tussling behavior, n=30 for each group, bars sharing the same letter are not significantly different according to Kruskal-Wallis test.

(C-E) The expression pattern of *P1^a-spGAL4* (C) and *pC1^{SS2}-spGAL4* (D) in male brain, and (E) the registration of *P1^a* (magenta) and *pC1^{SS2}* (green) neurons in a standard brain. The GFP expression indicates *P1^a* (magenta) and *pC1^{SS2}* (green) neurons in the male brain counterstained with nc82 (blue). White dashed boxes in (C-E) were zoomed in and shown below each panel. Scale bars, 100µm.

(F-H) Statistics on tussling (F), lunging (G) and unilateral wing extension (UWE, H) by *UAS-CsChrimson/+* males. Red light activation with a light intensity of 0.02 mW/mm² for 1 minute was applied at the 2nd and 5th minutes of this and subsequent experiments. n=32 for each group, n.s., not significant, Friedman test.

(I-K) Statistics on tussling (I), lunging (J) and UWE (K) in *P1^a>UAS-CsChrimson* males. n=32 for each group, n.s., not significant, ****P < 0.0001, Friedman test.

(L-N) Statistics on tussling (L), lunging (M) and UWE (N) in *pC1^{SS2}>UAS-CsChrimson* males. n=32 for each group, n.s., not significant, ***P < 0.001, ****P < 0.0001, Friedman test.

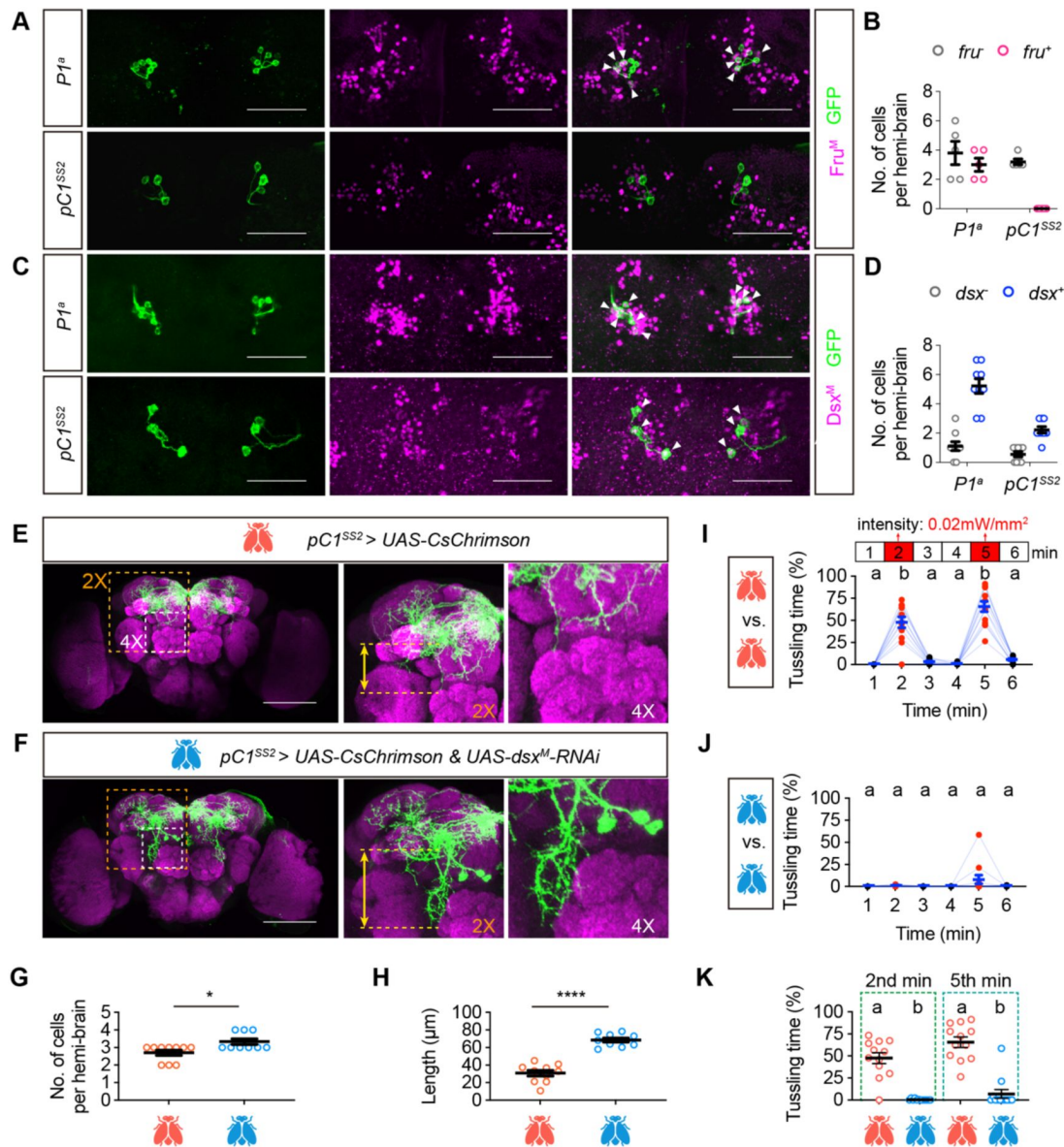


Figure 5.

Dsx^M regulates the development and tussling-promoting function of pC1^{SS2} neurons.

(A, B) The co-staining of *P1^a* or *pC1^{SS2}* neurons with the Fru^M antibody (A), and the quantification of *fru*-positive neurons in *P1^a* and *pC1^{SS2}* populations (B). n=5 for each group. Scale bars, 50µm.

(C, D) The co-staining of *P1^a* or *pC1^{SS2}* neurons with the Dsx^M antibody, and The quantification of *dsx*-positive neurons in *P1^a* and *pC1^{SS2}* populations (D), n=9 for each group. Scale bars, 50µm.

(E-K) Knocking down *dsx* in *pC1^{SS2}* neurons impact their development and tussling-promoting function. (E, F) Immunostaining of *pC1^{SS2}* neurons in control (E) and *dsx* knocking-down (F) males. Magenta represents nc82 signals, and green represents GFP signals. The images within the yellow dashed box and white dashed box are magnified 2× and 4×, respectively, and displayed on the right. Scale bars, 100µm. (G) Quantification of the number of *pC1^{SS2}* neurons in control and *dsx* knocking-down males. n=10, 9 for control and experimental group respectively, *P < 0.05, Mann-Whitney test. Error bars correspond to SEM. (H) Quantification of the neuronal projection length shown in the 2× magnified images in (E) and (F). n=10 for each group, ****P < 0.0001, Mann-Whitney test. (I-K) Tussling time by *pC1^{SS2} > UAS-CsChrimson* (I) and *pC1^{SS2} > UAS-CsChrimson & UAS-dsx^M-RNAi* (J) males, which were quantified in (K). Red light activation with a light intensity of 0.02 mW/mm² for 1 minute was applied at the 2nd and 5th minutes. n=12, bars sharing the same letter are not significantly different. Friedman test for (I) and (J), Kruskal-Wallis test for (K).

(**Figures 5E** [↗](#)–**5G** [↗](#)), and significantly lengthened the axonal projections ventrally (**Figure 5H** [↗](#)). We reasoned that these developmental abnormality may effect the tussling-promoting functions of pC1^{SS2} neurons, and tried to activate these neurons with or without knocking down *dsx*. We found that males with wild-type *dsx* function displayed intensive tussling behavior for approximately 50% and 70% of the total activation time, respectively, during the first and second activation periods, with the response returning to baseline levels after the removal of red light (**Figure 5I** [↗](#), **5K** [↗](#)). However, males with *dsx* knocked down in pC1^{SS2} neurons showed no tussling behavior during the first activation session, and few tussling behavior during the second activation session (**Figure 5J** [↗](#), **5K** [↗](#)). Together, these results reveal the crucial role of the *dsx* gene in the development and tussling-promoting function of pC1^{SS2} neurons.

Territory and Mating Advantage in Tussling-favored Experienced Males

As our results showed that GH males and SH males were more likely to perform the low-frequency, high-intensity tussling and the high-frequency, low intensity lunging, respectively, we wondered which fighting strategy was more practical in the competition for territory and mating resources.

To address this question, we first designed an experimental paradigm for territory competition (**Figure 6A** [↗](#)). A circular territory, which is provided with food and a fixed virgin female and separated by a wall with a small door open to the outer area, was competed by two tester males, one of which was marked on their wings. We manually analyzed the first 10 winning events, each was defined as one male successfully displacing the other from the inner resource area. The winning index was calculated as the proportional difference in the number of wins between the marked male and unmarked male. In control groups such as two GH males of the same age, we observed some ‘winner takes all’ phenomenon (one male won nearly all of the 10 encounters), but overall, the winning index was close to zero due to the randomness of marking (see methods). We further found that while GH 7-day-old males (G7) won slightly, but not significantly, more than SH 7-day-old males (S7) (**Figure 6B** [↗](#)), older G14 males had a significant advantage over the S14 males (**Figure 6C** [↗](#) and **Figure 6-figure supplement 1** [↗](#), **Table S2** [↗](#), see Figure 6-video supplement 1). Additionally, we observed that when G14 males were paired with S14 males, the proportion of males performing tussling behavior was not significantly different from that between two G14 males. However, the duration of each tussling event in G14 and S14 male pairs was significantly shorter than that between two G14 males (**Figure 6-figure supplement 2** [↗](#)).

Next, we conducted further investigations into the role of social experience and the two types of fighting behaviors in mating competition (**Figure 6D** [↗](#)). Two males and one virgin female were loaded into the behavioral chamber at the same time and recorded for courtship behaviors. A winner was defined as the male that first copulated with the female. We used the copulation advance index, which was calculated as the proportional difference between marked males and unmarked males in winning copulation (see methods). We found that G7 males had no significant advantage over S7 males (**Figure 6E** [↗](#)); however, G14 males displayed a significantly increased mating advantage over S14 males (**Figure 6F** [↗](#), see Figure 6-video supplement 2). Together, these results suggest that the tussling-favored GH males are more competitive in territory control and mating compared to the lunging-favored SH males.

The above results showed that GH males were more competitive compare to SH males, especially in older males, suggesting that social experience may play a potential role in altering age-related mating competition. To test this hypothesis, we first tested mating competition between males of different ages under the same rearing conditions. The results showed that younger males, whether reared in group or isolation, had a greater mating advantage compared to older males (G7 > G14, S7 > S14, G14 > G21, S14 > S21) (**Figure 6G** [↗](#), **Table S2** [↗](#)). These results indicate that there is generally a disadvantage in mating competition associated with aging. We next investigated

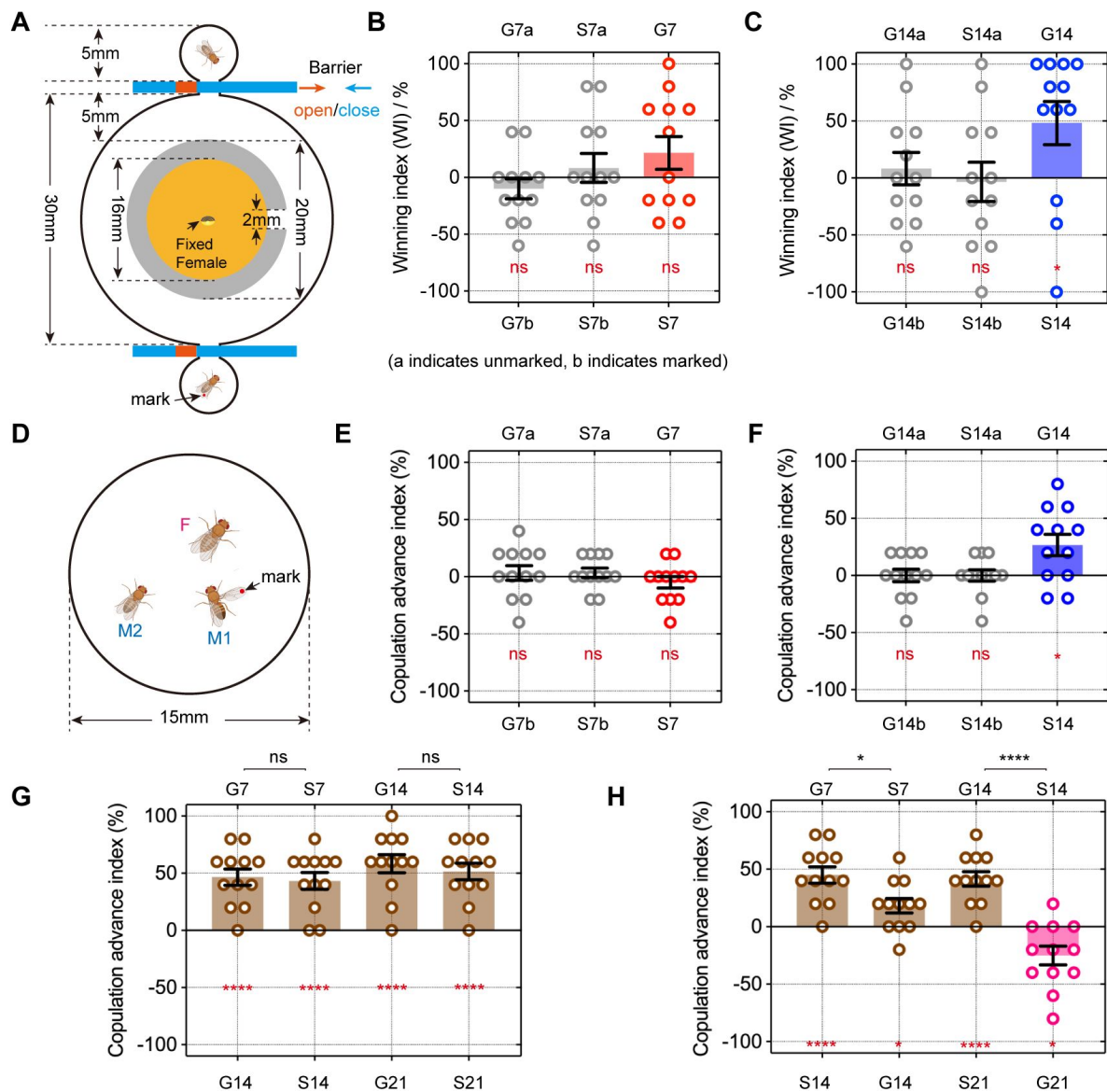


Figure 6.

Social enrichment enhances male territorial control and mating competition.

(A) Schematic diagram of the paradigm for testing territorial control in males.

(B) The territorial control test of 7-day-old GH (G7) and SH (S7) males. G7a and G7b stand for marked G7 male and unmarked G7 male, respectively. n=12 for each group, n.s., not significant, one sample t test.

(C) The territorial control test of 14-day-old GH (G14) and SH (S14) males. n=12 for each group, n.s., not significant, *P < 0.05, one sample t test.

(D) Schematic diagram of the paradigm for testing male mating advantage.

(E) The mating competition test between G7 and S7 males. n=12 for each group, n.s., not significant, one sample t test.

(F) The mating competition test between G14 and S14 males. n=12 for each group, n.s., not significant, *P < 0.05, one sample t test.

(G and H) The mating competition test between young males and old males, n=12 for each group, n.s., not significant, *P < 0.05, ****P < 0.0001, one sample t test for comparison within group (red labeling), Mann-Whitney test for comparison between groups (black labeling). Error bars correspond to SEM.

whether the mating advantage induced by social experience could compensate for the mating disadvantage caused by aging. We tested pairs of males with different ages and rearing conditions (G7 vs. S14, S7 vs. G14, G14 vs. S21, S14 vs. G21) for mating competition. We found that while 14-day-old males were less competitive than 7-day-old males regardless of the rearing condition, indicative of the aging-related disadvantage, 21-day-old GH males were significantly more competitive than 14-day-old SH males ($G21 > S14$) (**Figure 6H** [↗](#)).

To further investigate whether the territorial control and mating competition advantage in GH males are related to their high tussling behavior, we paired males with low tussling behavior (*Or47b-GAL4/UAS-Kir2.1* and *pC1^{SS2}/UAS-Kir2.1*) with control males for territorial control and mating competition tests. We found that males with inactivated Or47b or pC1^{SS2} neurons were incompetent for both territorial control and mating competition compared to control males (**Figure 6-figure supplement 3** [↗](#)), further supporting a correlation between the occurrence of tussling behavior and the acquisition of territorial and mating competition advantages.

Taken together, these results indicate that long-term social enrichment in older males increases their territorial and mating competition, which can overcome their mating disadvantage associated with aging (**Figure 7** [↗](#)).

Discussion

Regulation of physiology and behavior by social experience is an important feature across animal species. Here, we found that social experience shapes the fighting strategies of male flies, in which GH males tend to reduce the low-intensity lunging and increase the high-intensity tussling, which enhances their territorial and mating competition. We further showed that the two forms of fighting are mediated by distinct sensory and central neurons. The complex implications of these findings are discussed below.

Previous studies have found that social isolation generally enhances aggression but decreases mating competition in animal models (Palavicino-Maggio and Sengupta, 2022 [↗](#); Sethi et al., 2019 [↗](#); Yadav et al., 2024 [↗](#)), presenting a paradox between social experience, aggression and reproductive success. Our results resolved this paradox by dissecting aggressive behaviors into the well-studied, frequently observed lunging behavior and the less frequent but high-intensity tussling behavior. We improved the behavioral paradigm for studying tussling behavior and revealed that social experience oppositely regulates lunging and tussling. It is important to note that the frequency of lunging decreases while the frequency of tussling increases from 7-day-old to 14-day-old males, suggesting a shift in fighting strategy with age. Our results further suggest that social experience may enhance male mating competition by shifting their fighting strategy to the low-frequency, high-intensity tussling, which may be more effective in combat, although direct evidence is still lacking. These results are consistent with a previous study showing a correlation between the establishment of male dominance and tussling behavior, but not lunging (Simon and Heberlein, 2020 [↗](#)). Interestingly, we found that the mating advantage gained through social enrichment can even offset the mating disadvantage associated with aging, further supporting the vital role of shifting fighting strategies in experienced, aged males. Note that a previous study found a mating advantage in 7-day-old males compared to 2-day-old males (Lin et al., 2016 [↗](#)), which may be due to different levels of sexual maturity, but not necessarily reflect the aging effect. These results separate the function of the two fighting forms and rectify our understanding of how social experiences regulate aggression and reproductive success.

Our results revealed distinct neural circuits underlying the two forms of fighting, spanning from primary sensory neurons to central regulatory neurons. Specifically, silencing the cVA-sensing Or67d neurons reduced male lunging but not tussling, whereas inactivating Or47b neurons impaired male tussling but not lunging. Notably, Or47b neurons respond to pheromones common

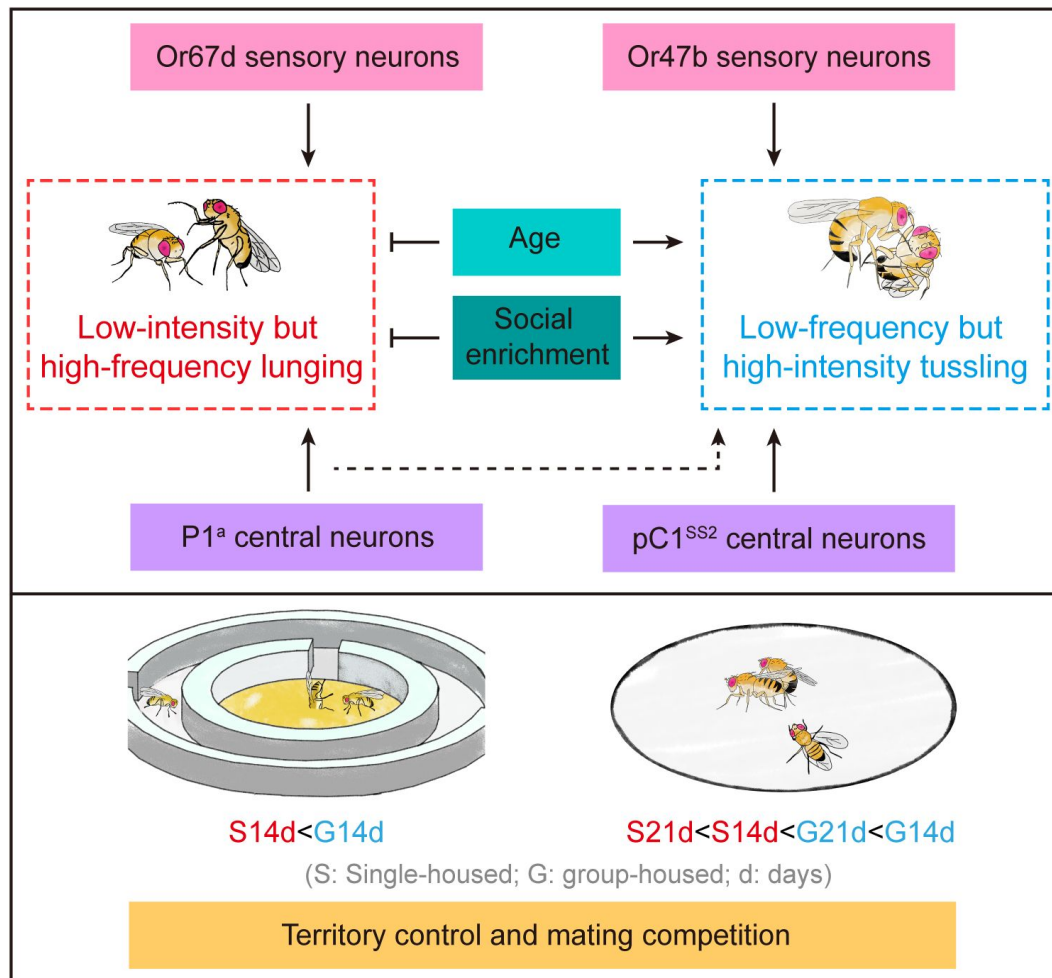


Figure 7.

Summary model of fighting strategies and reproductive success in male *Drosophila*.

Age and social enrichment inhibit the low-intensity, high-frequency lunging while promoting the low-frequency, high-intensity tussling. This shift in fighting strategies among experienced, aged males can enhance their territorial control and mating competition, even offsetting aging-related mating disadvantages. Lunging and tussling behaviors are regulated by distinct sensory and central neurons.

to both sexes, and the sensitivity of these sensory neurons is enhanced by social enrichment (Lin et al., 2016; Peng et al., 2021; Sethi et al., 2019). Unlike the Or67d-dependent lunging, which is primarily male-induced and socially inhibited, the Or47b-dependent tussling is promoted by social enrichment with either male or female flies. Recently, it has been found that female flies increase aggression towards mating pairs or mated females, depending on Or47b and Or67d, respectively (Gaspar et al., 2022; Wan et al., 2023). These results suggest a common function of Or47b and Or67d neurons in both sexes for regulating aggression. However, whether these neurons mediate different fighting patterns in females and how they respond to social experience in females remain unknown. Nevertheless, our results on these two sensory pathways in males provide crucial evidence for dissecting their functions.

Regarding central regulatory neurons, we identified a novel class of pC1 neurons, pC1^{SS2}, that specifically promote male tussling. In male flies, there are ~60 pairs of *dsx*-expressing pC1 neurons, including the well-characterized P1^a neurons that also express *fru*^M (Asahina, 2018; Jiang and Pan, 2022); however, there are only three pairs of pC1^{SS2} neurons that specifically express *dsx* but not *fru*^M, and are crucial for the high-intensity tussling in males. The three pairs of pC1^{SS2} neurons may have distinct function from P1^a neurons in several ways. (1) P1^a neurons promotes both male courtship and aggression (Hoopfer et al., 2015; Zhao et al., 2024), whereas pC1^{SS2} neurons specifically promote aggression but not courtship. A recent study also identified pC1^{SS2} neurons as aggression-promoting, but their roles in different fighting modes are not depicted (Hindmarsh Sten et al., 2025). That tussling-promoting pC1^{SS2} neurons specifically express *dsx* but not *fru*^M is also consistent with a previous finding that *dsx*-positive pC1 neurons, rather than *fru*^M-positive pC1 neurons, are crucial for male aggression (Koganezawa et al., 2016). (2) P1^a neurons promotes both lunging and tussling, although their function on tussling requires additional visual stimulation. In contrast, pC1^{SS2} neurons specifically promote tussling, and the presence of light enhances but is not absolutely necessary for tussling. These results align with previous findings that visual stimuli amplify P1 activation and related behaviors (Hindmarsh Sten et al., 2021; Kohatsu and Yamamoto, 2015; Pan et al., 2012), and visual stimulation is important for male aggression (Duistermars et al., 2018; Ramin et al., 2014; Schretter et al., 2020). (3) Male lunging is induced after P1^a activation; whereas male tussling is induced during pC1^{SS2} activation. These results suggest that P1^a neurons may not directly regulate aggression but instead induce a persistent state that promote aggression in general, while pC1^{SS2} neurons may play a more direct role in mediating tussling. Future studies are needed to further dissect the seemingly parallel sensory and central neuronal pathways involved in lunging and tussling, and to explore how these pathways interact during social experience and aging to optimize reproductive success.

Materials and Methods

Fly Stocks

Flies were maintained at 22 or 25 °C in a 12 h:12 h light: dark cycle. *Canton-S* flies were used as the wild-type strain and obtained from Janelia Research Campus (Pan and Baker, 2014). *Or67d-GAL4* (BDSC_9998), *Or65a-GAL4* (BDSC_9993), *Or47b-GAL4* (BDSC_9984), *IR84a-GAL4* (BDSC_41734), *TK-GAL4* (BDSC_51795), *R19B03-GAL4* (BDSC_49830), *UAS-dTrpA1* (BDSC_26263), *UAS-Kir2.1* (BDSC_6595), *UAS-CsChrimson* (BDSC_82181), *UAS-myrGFP* (BDSC_32198), *retro-Tango* (BDSC_99661) and *trans-Tango* (BDSC_77124) flies were obtained from Bloomington *Drosophila* Stock Center (BDSC). *UAS-NaChBac* (Ren et al., 2001), *fru*^{GAL4} (Stockinger et al., 2005), *P1^a-spGAL4* (Zhang et al., 2018), *pC1^{SS1}-spGAL4* (Wang et al., 2021), *pC1^{SS2}-spGAL4* (Wang et al., 2021), *UAS-dsx^M-RNAi* (Peng et al., 2022) and *UAS-fruMi* (Meissner et al., 2016) were used as previously described. *UAS-Or47b RNAi* (THU_2599) was obtained from TsingHua Fly Center. Detailed genotypes of flies were listed in **Table S1**.

Male aggression assay

For the aggression assays presented in **Figure 1**, the prepared food was evenly spread into the center of behavioral chambers. There was no female in the traditional paradigm (**Figure 1A**, left) (Wu et al., 2020; Zhou et al., 2008), while a 3-5 day old virgin female was gently fixed in the center of the food with its abdomen outside in the modified paradigm (**Figure 1A**, right). We used a fixed female to restrict its position in the center of food. Then two male testers were placed into the upper chambers separated from the food (with or without a fixed female) by a film barrier. All testers were loaded by cold anesthesia. After a 30-minute adaptation, the film was gently removed to allow the two males to fall into the behavioral chamber, and the aggressive behavior was recorded for 2 hours. The aggression assays in **Figure 1-figure supplement 1** was performed by using the paradigm without female, while other data were acquired using the modified paradigm with a fixed female unless otherwise stated. In some experiments, both single-housed and group-housed (30-40 males per vial) males were assayed for comparison.

We manually analyzed lunging and tussling behaviors in this study. Lunging is characterized by a male raising its forelegs and quickly striking the opponent, and each lunge typically lasts less than 0.2 seconds through detailed slow-motion analysis. Tussling is characterized by both males using their forelegs and bodies to tumble over each other, and this behavior may last from seconds to minutes. Tussling is often mixed with boxing, in which both flies rear up and strike the opponent with forelegs. Since boxing is often transient and difficult to distinguish from tussling, we referred to the mixed boxing and tussling behavior simply as tussling. As we manually analyze tussling for 2 hours for each pair of males, it is possible that we may miss some tussling events, especially those quick ones.

For the quantification of lunge latency and frequency in **Figures 2E**, **2F**, **3C**, **3D**, and **Figure 1-figures supplement 1A**, **1B**, we manually analyzed the videos and defined the lunge latency as the time point at which the first lunging event occurred between the two male flies after the start of the recording. We then counted the number of lunge events occurring within 10 minutes after the first lunge. For the quantification of tussle latency and frequency in **Figures 1B**, **1C**, **1D**, **2C**, **2D**, **3A**, **3B**, **3E**, **3F**, **3G**, **3H**, **4A** and **4D**, we manually analyzed the videos and defined the tussle latency as the time point at which the first tussling event occurred between the two male flies after the start of the recording. We defined the tussle frequency (No. of tussles in 2 hr) as the total number of tussling events that occurred within two hours from the beginning of the video.

The food recipes for the male aggression assay were listed below. A mixture of 2.5% yeast, 2.5% sucrose and 1% agar dissolved in apple juice (Huiyuan Apple juice concentrate) was used for **Figure 1B**; a mixture of 2.5% yeast, different concentrations of sucrose (0, 2.5%, 5% or 10%) and 1% agar dissolved in apple juice was used for **Figure 1C**; a mixture of different concentrations of yeast (0, 2.5%, 5% or 10%), 2.5% sucrose and 1% agar dissolved in apple juice was used for **Figure 1D**; A mixture of 10% yeast and 1% agar dissolved in apple juice was used for other experiments.

dTrpA1-mediated activation experiments

For thermogenetic activation assay showed in **Figure 4-figure supplement 1**, two 5-7 days old males were loaded into the circular chamber (d=15mm) under cold anesthesia and separated with a transparent film. Before the recording start, the chamber was transferred to the recording platform at a specific temperature (22°C or 30°C) for 30 minutes adaptation. After that, the film was removed, and the behavior was recorded for 1 hour. The 10-minute video after film extraction was used for behavioral analysis.

CsChrimson-mediated optogenetic activation experiments

For optogenetic activation assay showed in **Figure 4** and **5**, males were raised at 25°C on standard fly food, collected after eclosion and housed in the dark for 5-7 days on food containing 0.4 mM all *trans*-Retinal (MFCD00001550, Sigma-Aldrich, St. Louis, MO). The specific light stimulation protocols are showed in the schematic (**Figure 4F**). In brief, Two 5-7 days old males were loaded into the circular chamber (d=10mm) under freeze anesthesia and separated with a transparent film in the dark. Then the chamber was moved on the plane light source (IR channel: 865nm, red-light channel: 630nm) for 15 mins adaptation. Before video recording, the IR light channel was switched on and adjusted to the appropriate intensity for recording. The film was removed before the recording start, with the red-light channel switched on (constant red light, 0.02 mW/mm²) during the 2nd and 5th minutes of the experiment. The statistics for the tussling, lunging, and unilateral wing extension (UWE) are as follows. For tussling time and UWE time, we used LifesongY software (website: <https://sourceforge.net/projects/lifesongy/>), which is written by Dr. Chao Guo based on LifesongX (Bernstein et al., 1992), to calculate the percentage of the total time these behaviors occurred per minute. Specifically, the "tussling time (%)" refers to the percentage of time during the observation period in which tussling behavior occurred, relative to the total observation time. Similarly, "UWE time (%)" represents the percentage of time during the observation period in which UWE occurred. For lunges, we counted the number of lunging occurrences per minute.

Territorial competition experiment

For the territorial competition experiment in **Figures 6A-6C**, **Figure 6-figure supplement 3A**, the behavioral setup was produced using a 3D printer (Bambu Lab A1), as illustrated in the schematic, with a transparent cover plate placed on the top. The preparation of food and female were the same as the above aggression paradigm. The female was fixed in the center of the food and two male testers were placed separately in the two small chambers on both sides. Before initiating the experiment, the two barriers were closed. After a 30-minute adaptation, the barriers were opened to allow the males access to the central large chamber, and were closed once the males went inside, and the territorial behavior was recorded for 2 hours. The first 10 encounters occurring after 1 hour of recording were used for statistical analysis. The winning event was defined as a male driving the other male out of the food area. To discriminate between tester males, about half of them were marked on one of their wings using a red marker pen 24 hours before the experiment. For experiments using males with different ages and/or housing conditions, we calculated the winning index as (num. of wins by males [upper panel] – num. of wins by males [lower panel])/10 encounters * 100%. For control experiments using males of the same age and housing condition, we calculated the winning index as (num. of wins by unmarked males – num. of wins by marked males)/10 encounters * 100%, which is roughly zero due to the randomness of marking.

Mating competition experiment

For the mating competition experiment shown in **Figure 6D-6H**, **Figure 6-figure supplement 3B**, the wings of males were marked 24 hours prior to the experiment the same as described above. During the experiment, a 3-5 days old virgin female was placed into the chamber under freeze anesthesia and covered with a transparent film. Then the unmarked males were loaded into the chamber in the same manner and also covered with a transparent film. Finally, the marked males were loaded. After 30 minutes adaptation, two films were removed to allow the three flies to interact, and the behavior was recorded for 1 hour. The chambers without successful copulation during the 1-hour period were excluded from further analysis. For each condition, we used 120 efficient chambers with 10 random chambers collectively as one sample. For experiments using males with different ages and/or housing conditions, the copulation advance index was defined as (num. of male winners [upper panel] – num. of male winners [lower

panel)]/10 chambers * 100%. For control experiments using males of the same age and housing condition, the copulation advance index was calculated as (num. of winners by unmarked males – num. of winners by marked males)/10 chambers * 100%, which is roughly zero due to the randomness of marking. The winner was defined as the male that first copulated with the female.

Quantitative Real-time PCR

Adult fly samples were frozen in liquid nitrogen and decapitated by vortex. The heads were then separated from the bodies using metal sieves. Each sample, consisting of 40 frozen heads, was used for total RNA preparation using TRIzol reagent (15596026, Thermo Fisher Scientific, Waltham, MA). We purified total RNA using a DNA-freeTM Kit (AM1906, Thermo Fisher Scientific, Waltham, MA) and performed reverse transcription using SuperScriptTM IV (18091050, Thermo Fisher Scientific, Waltham, MA) to obtain cDNA used for templates. Quantitative PCR was performed on the Roche LightCycler[®] 96 Real-Time PCR machine using AceQ qPCR SYBR Green Master Mix (Q121-02, Vazyme, Nanjing). Transcript levels were analyzed by the 2^{-ΔCT} method using *actin* as control. Each sample was run in triplicate. Each experiment was repeated three times using independent sets of genetic crosses. Primers used for RT-qPCR quantification were:

actin forward: 5'-GTCGCGATTAAACCGACTACCTGA-3'

actin reverse: 5'-TCTTGCTT CGAGATCCACATCTGC-3'

fru-P1 forward: 5'-GTGTGCGTACGTTTGAGTGT-3'

fru-P1 reverse: 5'-TAATCCTGTGACGTGCCAT-3'

Or47b forward: 5'-CAAATCTCAGCCTTCTGCGG-3'

Or47b reverse: 5'-GATACTGGCACAGCAAACCTCA-3'

Tissue Dissection, Staining, and Imaging

Brains were dissected in Schneider's insect medium (S2) and fixed in 4% paraformaldehyde in 0.5% Triton X-100 and 0.5% bovine serum albumin in phosphate-buffered saline (PAT) for 30 min at room temperature. After 4 × 10-min washes, tissues were blocked in 3% normal goat serum (NGS) for 60 min, then incubated in primary antibodies diluted in 3% NGS for 4 h at room temperature and 1 day at 4°C, then washed in PAT, and incubated in secondary antibodies diluted in 3% NGS for 4 h at room temperature and 1 day at 4°C. The tissues were then washed thoroughly in PAT and mounted for imaging. Primary antibodies used in this study include mouse anti-HA (BioLegend, 901501, 1:200), rabbit anti-GFP (Invitrogen, A11122, 1:1000), monoclonal mouse anti-GFP (Sigma-Aldrich, G6539, 1:500), mouse anti-Bruchpilot (Developmental Studies Hybridoma Bank, nc82, 1:50), rabbit anti-Dsx^M (1:500) and mouse anti-Fru^M (1:2000). Secondary antibodies used include donkey anti-mouse IgG conjugated to Alexa 488 (Invitrogen, A21206, 1:500), donkey anti-mouse IgG conjugated to Alexa 555 (Invitrogen, A-31570, 1:500), donkey anti-rabbit IgG conjugated to Alexa 488 (Invitrogen, A-21206, 1:500) and donkey anti-rabbit IgG conjugated to Alexa 555 (Invitrogen, A-31572, 1:500). Samples were imaged on Zeiss 900 confocal microscopes using ZEN and processed with Fiji software.

Trans-synaptic labeling by *trans*-Tango and *retro*-Tango

The *trans*-Tango or *retro*-Tango ligand with GFP was expressed under the GAL4 driver. Trans-synaptic signals were marked by a membrane-associated form of tdTomato (QUAS-mtdTomato-3xHA). The anti-HA antibody (BioLegend, 901501, 1:200) was applied to characterize the postsynaptic or presynaptic signal. For **Figure 3-figure supplement 2** [↗](#), male testers were kept at

18 °C until eclosion and subsequently single- or group-housed at the same temperature for two weeks before dissection and imaging. For **Figure 4-figure supplement 2**, flies were kept at 18 °C for *trans*-Tango and 25 °C for *retro*-Tango experiments.

Brain image registration

The protocol for standard brain registration is described previously (Zhou et al., 2014). Confocal images for brains containing P1^a neurons and pC1^{SS2} neurons were registered onto a standard reference brain (Zhou et al., 2014), using a Fiji graphical user interface as described previously (Ostrovsky et al., 2013).

Statistics

Experimental flies and genetic controls were tested under the same conditions, and data were collected from at least two independent experiments. Statistical analysis was performed using GraphPad Prism 9 (<https://www.graphpad.com/scientific-software/prism/>) as indicated in each figure legend. Data were first verified for normal distribution by the D'Agostino–Pearson normality test. For normally-distributed data, Student's *t* test was used for pairwise comparisons, and one-way ANOVA was used for comparisons among multiple groups, followed by Tukey's multiple comparisons. For data not normally-distributed, the Mann-Whitney test was used for pairwise comparisons, and the Kruskal–Wallis test was used for comparison among multiple groups, followed by Dunn's multiple comparisons. For group matched data that were normally-distributed, the Paired *t* test signed rank test was used for pairwise comparisons, and RM one-way ANOVA was used for comparison among multiple groups, followed by RM one-way ANOVA multiple comparisons. For group matched data that were not normally-distributed, the Wilcoxon matched-pairs signed rank test was used for pairwise comparisons, and the Friedman test was used for comparison among multiple groups, followed by Dunn's multiple comparisons. For qPCR experiments, the average relative expression of three independent experiments was analyzed using Mann-Whitney test. For one group compared with zero, we used one sample *t* test. The Chi-square test was used to compare male tussling percentage between two groups.

Figures and figure legends

Additional files

Figure 1-video supplement 1. Tussling behavior between two 14-day-old group-housed (G14) wild-type males. [🔗](#)

Figure 4-video supplement 1. Optogenetic activation of pC1^{SS2} neurons acutely induces tussling in 5 to 7-day-old group-housed males (pC1^{SS2}>UAS-CsChrimson). [🔗](#)

Figure 6-video supplement 1. In a territory competition paradigm, a 14-day-old group-housed (G14, marked) wild-type male consistently outcompetes a 14-day-old single-housed (S14, unmarked) wild-type male. [🔗](#)

Figure 6-video supplement 2. In a mating competition paradigm, a 14-day-old group-housed (G14, unmarked) wild-type male outperformed a 14-day-old single-housed (S14, marked) wild-type male and successfully copulated with the female. [🔗](#)

Figure 1-figure supplement 1.

Aging suppresses lunging behavior in male *Drosophila*.

(A) The lunge latency in 7-day-old single-housed (S7), 14-day-old single-housed (S14), 7-day-old group-housed (G7) and 14-day-old group-housed (G14) males, $n=12$ for each group, n.s., not significant, $*P < 0.05$, Mann-Whitney test. (B) The number of lunges in S7, S14, G7 and G14 males within 10 minutes, $n=12$ for each group, $*P < 0.05$, $****P < 0.0001$, Mann-Whitney test. Error bars correspond to SEM.

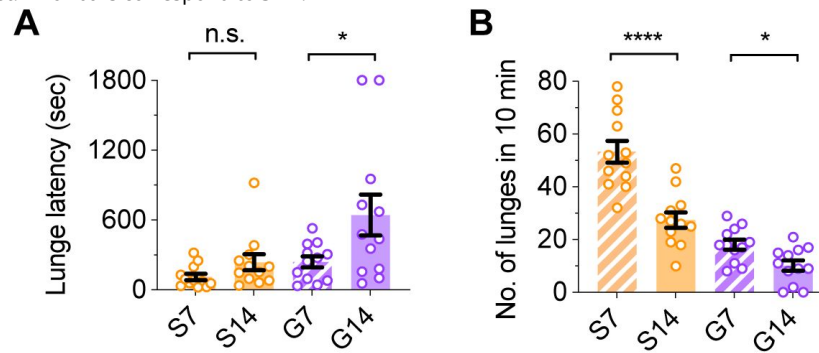


Figure 3-figure supplement 1.

Validation of the *Or47b* and *fru^M* RNAi efficiency.

Validation of *UAS-Or47b-RNAi* (A) and *UAS-fru^M* (B) efficiency by quantitative qPCR. $n=9$ for each of three replicates, $****P < 0.0001$, $***P < 0.001$, Mann-Whitney test. Error bars correspond to SEM.

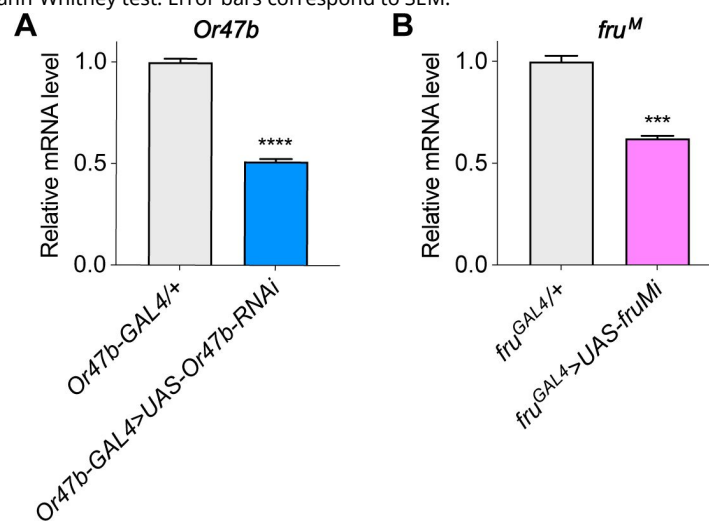


Figure 3-figure supplement 2.

Group housing increases downstream signals of *Or47b* ORNs.

(A) Projection of *Or47b* ORNs (green) and their downstream neurons (red) in GH and SH conditions. Scale bar, 100µm. (B) The fluorescence intensity of neurons downstream of *Or47b* ORNs is stronger under GH conditions compared to SH conditions. Both groups of male testers were 14 days old, n=10 and 8, respectively, ****P < 0.0001, Mann-Whitney test. Error bars correspond to SEM.

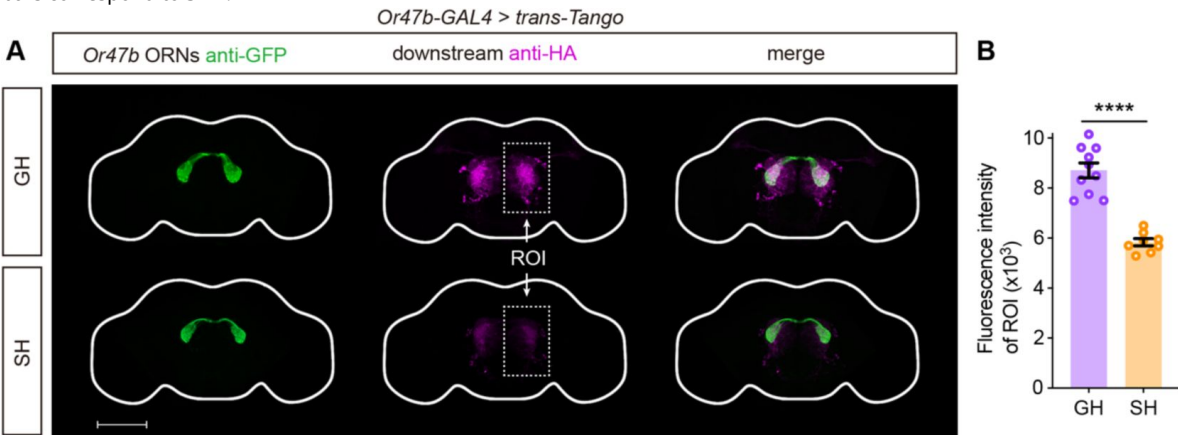
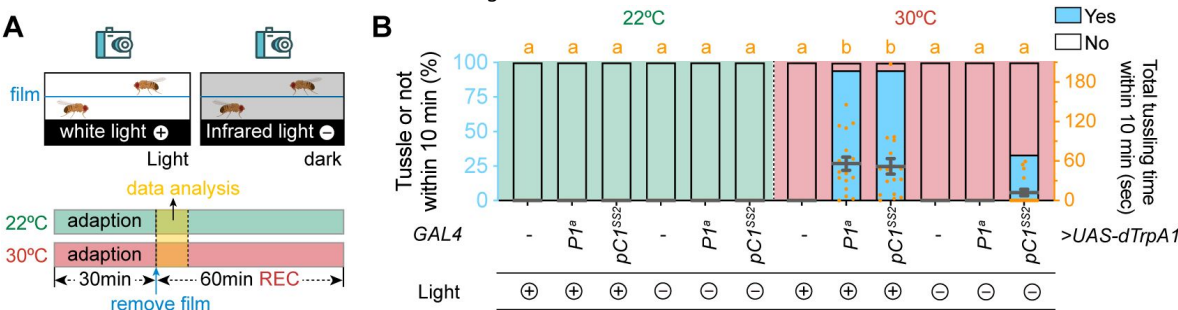


Figure 4-figure supplement 1.

Light-dependent initiation of male tussling behavior.

(A) Schematic diagram of the thermal activation experiment. (B) The percentage of tussling occurrence (left) and total tussling time (right) within 10 mins. All tester males were 5 to 7-day-old group-housed males. n=18 for each group, bars sharing the same letter are not significantly different according to Kruskal-Wallis test based on the data of total tussling time.



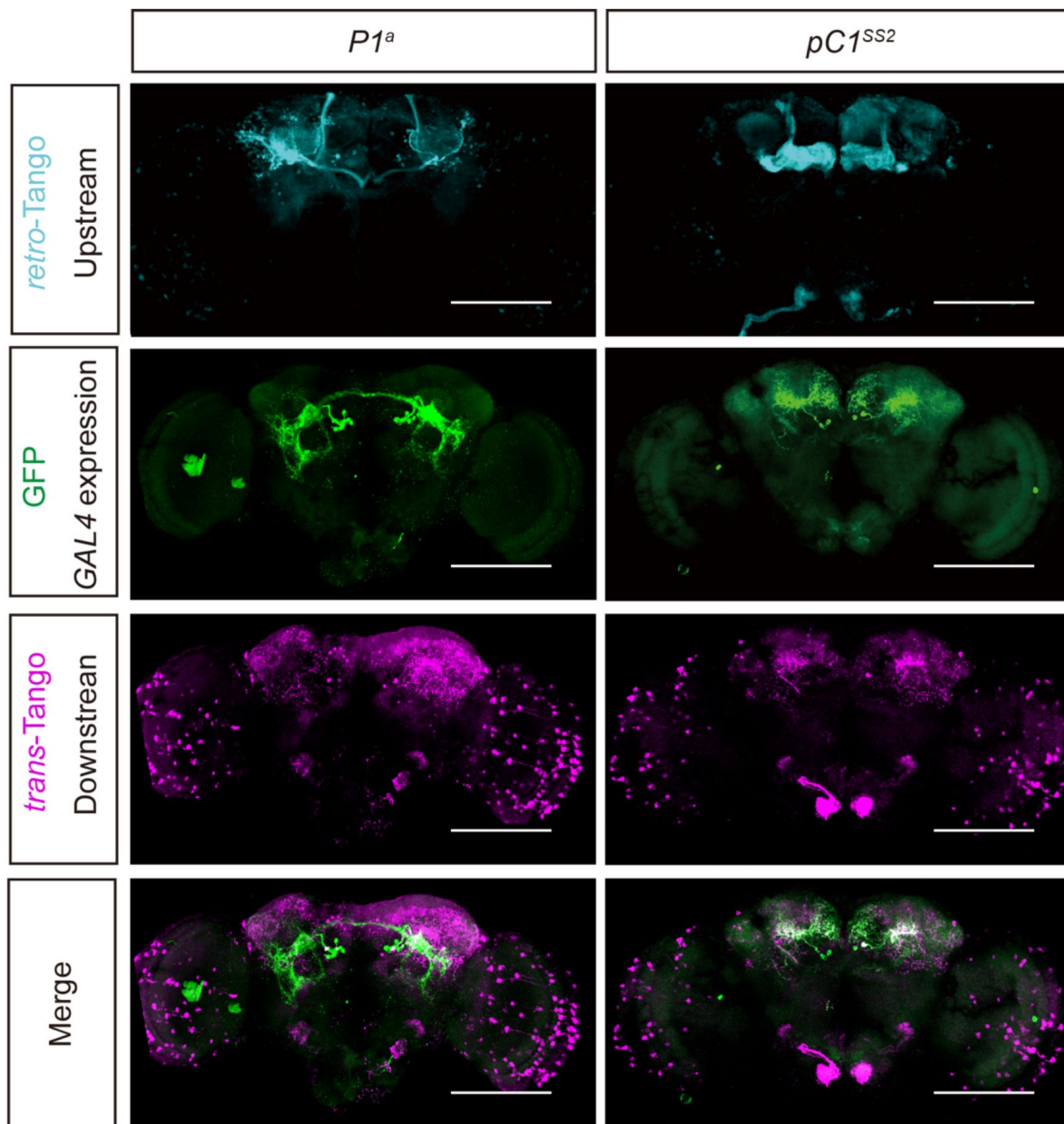


Figure 4-figure supplement 2.

Immunostaining images of potential upstream signals (*retro-Tango*) and downstream signals (*trans-Tango*) of *P1^a* and *pC1^{SS2}* neurons.

Scale bars, 100μm.

Figure 6-figure supplement 1.

Winning events of representative samples for territorial competition.

The blue and red bars represent the wins for S14 males and G14 males, respectively. An example formula is provided for calculating the winning index in the dashed box.

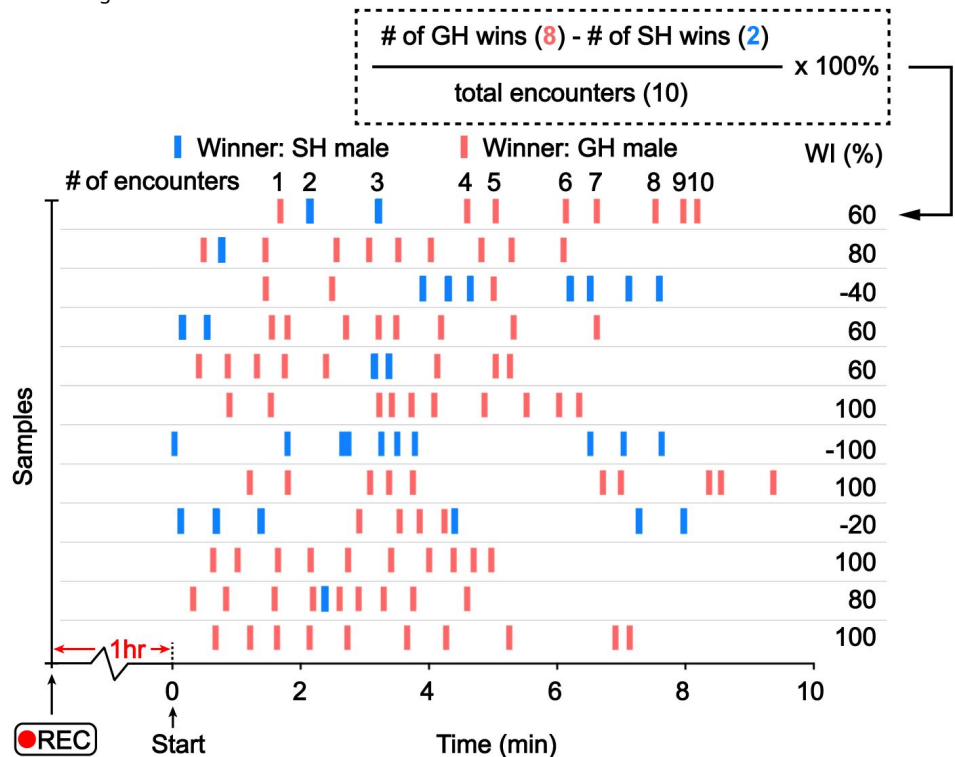
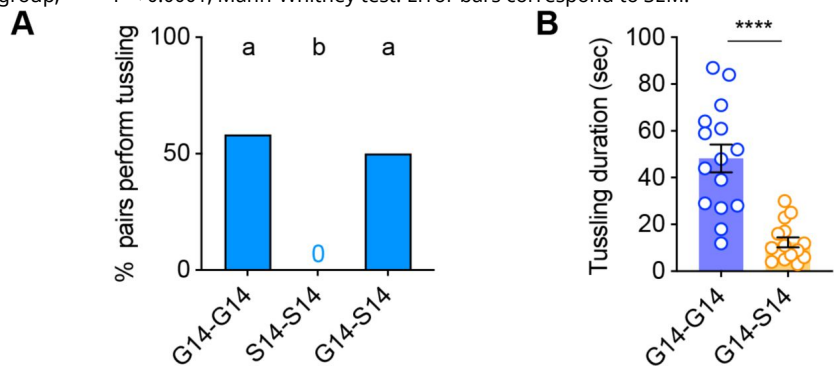


Figure 6-figure supplement 2.

The occurrence and duration of tussling behavior in male flies under different pairing conditions.

(A) Proportion of male pairs performing tussling within 2 hours under three different pairing conditions in the territorial control paradigm. $n=12$ for each group, bars sharing the same letter are not significantly different according to Chi-square test.

(B) Duration of each tussling event in males under the G14-G14 and G14-S14 pairing conditions. $n=15$ events from 8-12 samples for each group, **** $P < 0.0001$, Mann-Whitney test. Error bars correspond to SEM.



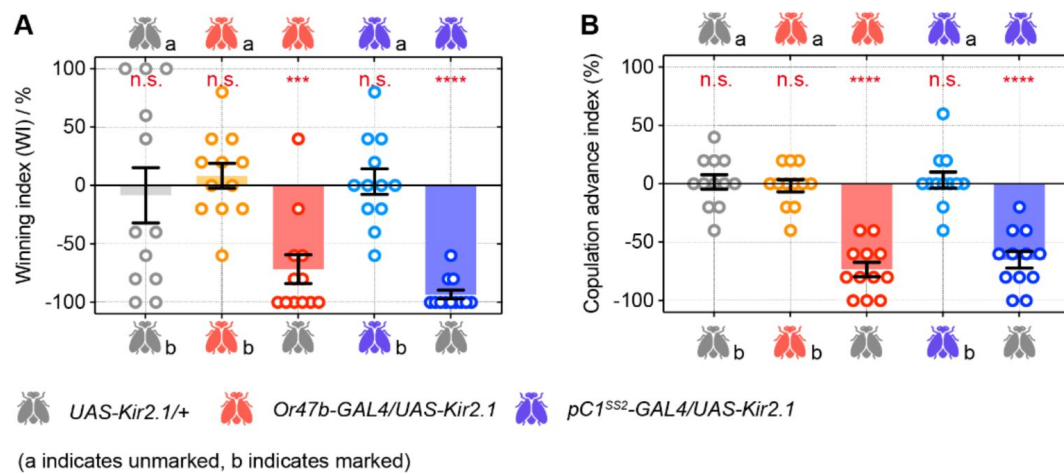


Figure 6-figure supplement 3.

Silencing *Or47b* or *pC1^{SS2}* neurons compromises male territorial control and mating competition.

(A, B) Effects of *Or47b* neurons inactivation and *pC1^{SS2}* neurons inactivation on territorial control (A) and mating competition (B) in male flies. n=12 for each group, n.s., not significant, ***P < 0.001, ****P < 0.0001, one sample t test.

Figures	Genotypes	Housing conditions	Sample sizes
Figure 1			
B	<i>Canton-S</i>	GH-2d, GH-7d, GH-14d	n=30, 30, 30, 46, 46, 46
C	<i>Canton-S</i>	GH-14d	n=30 for each group
D	<i>Canton-S</i>	GH-14d	n=30 for each group
Figure 2			
B	<i>Canton-S</i>	SH-14d, GH-14d	
C, D, E, F	<i>Canton-S</i>	SH-14d, GH-14d	n=27 for SH, n=30 for GH
G	<i>Canton-S</i>	SH-14d, GH-14d	n=30 events from 10 samples for each group
Figure 3			
A, B, C, D	<i>w+;;UAS-Kir2.1/+</i>	GH-14d	30
	<i>w+;;Or67d-GAL4/UAS-Kir2.1</i>	GH-14d	30
	<i>w+;;Or65a-GAL4/+;UAS-Kir2.1/+</i>	GH-14d	30
	<i>w+;;Or47b-GAL4/UAS-Kir2.1</i>	GH-14d	30
	<i>w+;;Ir84a-GAL4/UAS-Kir2.1</i>	GH-14d	30
E, F	<i>w+;;Or47b-GAL4/+</i>	GH-14d	30
	<i>w+;;Or47b-GAL4/UAS-Or47b-RNAi</i>	GH-14d	30
	<i>w+;;Or47b-GAL4/UAS-fruMi</i>	GH-14d	30
G, H	<i>w+;;Or47b-GAL4/+</i>	GH-14d	60
	<i>w+;;Or47b-GAL4/UAS-NaChBac</i>	GH-14d	60
	<i>w+;;Or47b-GAL4/+</i>	SH-14d	60
	<i>w+;;Or47b-GAL4/UAS-NaChBac</i>	SH-14d	49
Figure 4			
A, B	<i>w+;;UAS-Kir2.1/+</i>	GH-14d	30
	<i>w+;;R19B03-GAL4/UAS-Kir2.1</i>	GH-14d	30
	<i>w+;;R15A01-AD/+;R71G01-DBD/UAS-Kir2.1 (P1^a > UAS-Kir2.1)</i>	GH-14d	30
	<i>w+;;TK-GAL4/UAS-Kir2.1 (GH-14d)</i>	GH-14d	30
	<i>w+;;VT002064-AD/+;VT008469DBD/UAS-Kir2.1 (pCI^{SS1} > UAS-Kir2.1)</i>	GH-14d	30
	<i>w+;;VT002064-AD,dxDBD/UAS-Kir2.1 (pCI^{SS2} > UAS-Kir2.1)</i>	GH-14d	30
C	<i>w-;;R15A01-AD/UAS-myrGFP; R71G01-DBD/+ (P1^a > UAS-GFP)</i>	GH-5-7d	
D	<i>w-;;UAS-myrGFP/+;VT002064-AD,dxDBD/+ (pCI^{SS2} > UAS-GFP)</i>	GH-5-7d	

Table S1.

Detailed information for fly stocks.

F, G, H	<i>w-;;UAS-CsChrimson/+</i>	GH-5-7d	32
I, J, K	<i>w-;;R15A01-AD/+;R71G01-DBD/UAS-CsChrimson (P1^a > UAS-CsChrimson)</i>	GH-5-7d	32
L, M, N	<i>w-;;VT002064-AD,dxDBD/UAS-CsChrimson (pC1^{SS2} > UAS-CsChrimson)</i>	GH-5-7d	32
Figure 5			
A, B, C, D	<i>w-;;R15A01-AD/UAS-myrGFP;R71G01-DBD/+ (P1^a > UAS-GFP)</i>	GH-5-7d	n=5 for each group (B)
	<i>w-;;UAS-myrGFP/+;VT002064-p65AD,dxDBD/+ (pC1^{SS2} > UAS-GFP)</i>	GH-5-7d	n=9 for each group (D)
E	<i>w-;;UAS-CsChrimson/+;VT002064-AD, dxDBD/+ (pC1^{SS2} > UAS-CsChrimson)</i>	GH-5-7d	
F	<i>w-;;UAS-CsChrimson/+; VT002064-AD, dxDBD/UAS-dsx^M-RNAi (pC1^{SS2} > UAS-CsChrimson & UAS-dsx^M-RNAi)</i>	GH-5-7d	
G	<i>w-;;UAS-CsChrimson/+;VT002064-AD, dxDBD/+ (pC1^{SS2} > UAS-CsChrimson)</i>	GH-5-7d	10
	<i>w-;;UAS-CsChrimson/+;VT002064-AD, dxDBD/UAS-dsx^M-RNAi (pC1^{SS2} > UAS-CsChrimson & UAS-dsx^M-RNAi)</i>	GH-5-7d	9
H	<i>w-;;UAS-CsChrimson/+;VT002064-AD, dxDBD/+ (pC1^{SS2} > UAS-CsChrimson)</i>	GH-5-7d	10
	<i>w-;;UAS-CsChrimson/+;VT002064-AD, dxDBD/UAS-dsx^M-RNAi (pC1^{SS2} > UAS-CsChrimson & UAS-dsx^M-RNAi)</i>	GH-5-7d	10
I	<i>w-;;UAS-CsChrimson/+;VT002064-AD, dxDBD/+ (pC1^{SS2} > UAS-CsChrimson)</i>	GH-5-7d	12
J	<i>w-;;UAS-CsChrimson/+;VT002064-AD, dxDBD/UAS-dsx^M-RNAi (pC1^{SS2} > UAS-CsChrimson & UAS-dsx^M-RNAi)</i>	GH-5-7d	12
K	<i>w-;;UAS-CsChrimson/+;VT002064-AD, dxDBD/+ (pC1^{SS2} > UAS-CsChrimson)</i>	GH-5-7d	12
	<i>w-;;UAS-CsChrimson/+;VT002064-AD, dxDBD/UAS-dsx^M-RNAi (pC1^{SS2} > UAS-CsChrimson & UAS-dsx^M-RNAi)</i>	GH-5-7d	12

Table S1. (continued)

	<i>w-;UAS-CsChrimson/+;VT002064-AD, dsxDBD/+ (pCI^{SS2} > UAS-CsChrimson)</i>	GH-5-7d	12
	<i>w-;UAS-CsChrimson/+;VT002064-AD, dsxDBD/UAS-dsx^M-RNAi (pCI^{SS2} > UAS-CsChrimson & UAS-dsx^M-RNAi)</i>	GH-5-7d	12
Figure 6			
B	<i>Canton-S</i>	GH-7d	12
	<i>Canton-S</i>	SH-7d	12
	<i>Canton-S</i>	GH-7d, SH-7d	12
C	<i>Canton-S</i>	GH-14d	12
	<i>Canton-S</i>	SH-14d	12
	<i>Canton-S</i>	GH-14d, SH-14d	12
E	<i>Canton-S</i>	GH-7d	12
	<i>Canton-S</i>	SH-7d	12
	<i>Canton-S</i>	GH-7d, SH-7d	12
F	<i>Canton-S</i>	GH-14d	12
	<i>Canton-S</i>	SH-14d	12
	<i>Canton-S</i>	GH-14d, SH-14d	12
G	<i>Canton-S</i>	GH-7d, GH-14d	12
	<i>Canton-S</i>	SH-7d, SH-14d	12
	<i>Canton-S</i>	GH-14d, GH-21d	12
	<i>Canton-S</i>	SH-14d, SH-21d	12
H	<i>Canton-S</i>	GH-7d, SH-14d	12
	<i>Canton-S</i>	SH-7d, GH-14d	12
	<i>Canton-S</i>	GH-14d, SH-21d	12
	<i>Canton-S</i>	SH-14d, GH-21d	12
Figure 1-figure supplement 1			
A, B	<i>Canton-S</i>	SH-7d	12
	<i>Canton-S</i>	SH-14d	12
	<i>Canton-S</i>	GH-7d	12
	<i>Canton-S</i>	GH-14d	12
Figure 3-figure supplement 1			
A	<i>w+;;Or47b-GAL4/+</i>	GH-5-7d	9
	<i>w+;;Or47b-GAL4/UAS-Or47b-RNAi</i>	GH-5-7d	9
B	<i>w+;;fru^{GAL4}/+</i>	GH-5-7d	9

Table S1. (continued)

	<i>w+;;fru^{GAL4}/UAS-fruMi</i>	GH-5-7d	9
Figure 3-figure supplement 2			
A	<i>UAS-myrGFP,QUAS-mtdtomato(3xHA);trans-Tango/+;Or47b-GAL4/+</i>	GH-14d, 18°C	
	<i>UAS-myrGFP,QUAS-mtdtomato(3xHA);trans-Tango/+;Or47b-GAL4/+</i>	SH-14d, 18°C	
B	<i>UAS-myrGFP,QUAS-mtdtomato(3xHA);trans-Tango/+;Or47b-GAL4/+</i>	GH-14d, 18°C	10
	<i>UAS-myrGFP,QUAS-mtdtomato(3xHA);trans-Tango/+;Or47b-GAL4/+</i>	SH-14d, 18°C	8
Figure 4-figure supplement 1			
B	<i>w+;UAS-dTrpA1/+ (light, 22°C)</i>	GH-5-7d	18
	<i>w+;R15A01-AD/UAS-dTrpA1;R71G01-DBD/+ (P1^a > UAS-dTrpA1) (light, 22°C)</i>	GH-5-7d	18
	<i>w-;UAS-dTrpA1/+; VT002064-AD, dsxDBD/+ (pCI^{SS2} > UAS-dTrpA1) (light, 22°C)</i>	GH-5-7d	18
	<i>w+;UAS-dTrpA1/+ (dark, 22°C)</i>	GH-5-7d	18
	<i>w+;R15A01-AD/UAS-dTrpA1;R71G01-DBD/+ (P1^a > UAS-dTrpA1) (dark, 22°C)</i>	GH-5-7d	18
	<i>w-;UAS-dTrpA1/+;VT002064-AD, dsxDBD/+ (pCI^{SS2} > UAS-dTrpA1) (dark, 22°C)</i>	GH-5-7d	18
	<i>w+;UAS-dTrpA1/+ (light, 30°C)</i>	GH-5-7d	18
	<i>w+;R15A01-AD/UAS-dTrpA1;R71G01-DBD/+ (P1^a > UAS-dTrpA1) (light, 30°C)</i>	GH-5-7d	18
	<i>w-;UAS-dTrpA1/+;VT002064-AD, dsxDBD/+ (pCI^{SS2} > UAS-dTrpA1) (light, 30°C)</i>	GH-5-7d	18
	<i>w+;UAS-dTrpA1/+ (dark, 30°C)</i>	GH-5-7d	18
	<i>w+;R15A01-AD/UAS-dTrpA1;R71G01-DBD/+ (P1^a > UAS-dTrpA1) (dark, 30°C)</i>	GH-5-7d	18
	<i>w-;UAS-dTrpA1/+;VT002064-AD, dsxDBD/+ (pCI^{SS2} > UAS-dTrpA1) (dark, 30°C)</i>	GH-5-7d	18
Figure 4-figure supplement 2			
retro-Tango	<i>QUAS-mtdTomato(3xHA);retro-Tango/R15A01-AD;UAS-EGFP/R71G01-DBD (P1^a > retro-Tango)</i>	GH-14-21d	

Table S1. (continued)

	<i>QUAS-mtdTomato(3xHA);retro-Tango/+;UAS-EGFP/VT002064AD,dxDBD</i> ($pCI^{SS2} > retro-Tango$)	GH-14-21d	
GFP	<i>w-;R15A01-AD/+;R71G01-DBD/UAS-myrGFP</i> ($P1^a > GFP$)	GH-5-7d	
	<i>w-;;VT002064AD,dxDBD/UAS-myrGFP</i> ($pCI^{SS2} > GFP$)	GH-5-7d	
trans-Tango	<i>UAS-myrGFP,QUAS-mtdtomato(3xHA);trans-Tango/R15A01-AD;R71G01-DBD/+</i> ($P1^a > trans-Tango$)	GH-14-21d	
	<i>UAS-myrGFP,QUAS-mtdtomato(3xHA);trans-Tango/+;VT002064AD,dxDBD/+</i> ($pCI^{SS2} > trans-Tango$)	GH-14-21d	
Figure 6-figure supplement 1			
	<i>Canton-S</i>	GH-14d, SH-14d	12
Figure 6-figure supplement 2			
A	<i>Canton-S</i>	GH-14d	12
	<i>Canton-S</i>	SH-14d	12
	<i>Canton-S</i>	GH-14d, SH-14d	12
B	<i>Canton-S</i>	GH-14d	n=15 events from 8-12 samples for each group
	<i>Canton-S</i>	GH-14d, SH-14d	n=15 events from 8-12 samples for each group
Figure 6-figure supplement 3			
A	<i>w+;;UAS-Kir2.1/+</i>	GH-14d	12
	<i>w+;;Or47b-GAL4/UAS-Kir2.1</i>	GH-14d	12
	<i>w+;;Or47b-GAL4/UAS-Kir2.1</i>	GH-14d	12
	<i>w+;;UAS-Kir2.1/+</i>		
	<i>w+;;VT002064AD,dxDBD/ UAS-Kir2.1</i> ($pCI^{SS2} > UAS-Kir2.1$)	GH-14d	12
	<i>w+;;VT002064AD, dxDBD/ UAS-Kir2.1</i> ($pCI^{SS2} > UAS-Kir2.1$)	GH-14d	12
B	<i>w+;;UAS-Kir2.1/+</i>		
	<i>w+;;UAS-Kir2.1/+</i>	GH-14d	12
	<i>w+;;Or47b-GAL4/UAS-Kir2.1</i>	GH-14d	12
	<i>w+;;Or47b-GAL4/UAS-Kir2.1</i>	GH-14d	12
	<i>w+;;UAS-Kir2.1/+</i>		
	<i>w+;;VT002064AD,dxDBD/ UAS-Kir2.1</i> ($pCI^{SS2} > UAS-Kir2.1$)	GH-14d	12

Table S1. (continued)

	<i>w+;;VT002064AD, dsxDBD/ UAS-Kir2.1(pC1^{SS2} > UAS-Kir2.1)</i> <i>w+;;UAS-Kir2.1/+</i>	GH-14d	12
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Table S1. (continued)

Territorial control assay				
	pairs	winning index	P value	conclusion
Fig. 6B	G7a vs G7b	-10±30.15	0.275	/
	S7a vs S7b	8.33±43.87	0.524	/
	G7 vs S7	21.66±50.06	0.1619	/
Fig. 6C	G14a vs G14b	8.33±49.33	0.5702	/
	S14a vs S14b	-3.33±59.60	0.8499	/
	G14 vs S14	48.33±65.76	0.0272	G14>S14
Mating competition assay				
	pairs	copulation advance index	P value	conclusion
Fig. 6E	G7a vs G7b	3.33±22.29	0.6147	/
	S7a vs S7b	3.33±14.35	0.4382	/
	G7 vs S7	-5±17.32	0.3388	/
Fig. 6F	G14a vs G14b	0±19.07	>0.9999	/
	S14a vs S14b	0±17.06	>0.9999	/
	G14 vs S14	26.67±32.29	0.0155	G14>S14
Fig. 6G	G7 vs G14	46.67±24.62	<0.0001	G7>G14
	S7 vs S14	43.33±25.35	<0.0001	S7>S14
	G14 vs G21	58.33±27.58	<0.0001	G14>G21
	S14 vs S21	51.67±24.80	<0.0001	S14>S21
Fig. 6H	G7 vs S14	45±24.31	<0.0001	G7>S14
	S7 vs G14	18.33±21.67	0.0137	S7>G14
	G14 vs S21	41.67±21.67	<0.0001	G14>S21
	S14 vs G21	-25±28.44	0.0112	S14<G21
S21<S14<G21<G14				

Table S2.

Detailed comparison for territorial control and mating competition.

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Reviewer #1 (Public review):

This work addresses an important question in the field of *Drosophila* aggression and mating. Prior social isolation is known to increase aggression in males, manifesting as increased lunging, which is suppressed by group housing (GH). However, it is also known that single housed (SH) males, despite their higher attempts to court females, are less successful. Here, Gao et al., develop a modified aggression assay to address this issue by recording aggression in *Drosophila* males for 2 hours, with a virgin female immobilized by burying its head in the food. They found that while SH males frequently lunge in this assay, GH males switch to higher intensity but very low frequency tussling. Constitutive neuronal silencing and activation experiments implicate cVA sensing Or67d neurons in promoting high frequency lunging, similar to earlier studies, whereas Or47b neurons promote low frequency but higher intensity tussling. Optogenetic activation revealed that three pairs of pC1SS2 neurons increase tussling. Cell-type-specific DsxM manipulations combined with morphological analysis of pC1SS2 neurons and side-by-side tussling quantification link the developmental role of DsxM to the functional output of these aggression-promoting cells. In contrast, although optogenetic activation of P1a neurons in the dark did not increase tussling, thermogenetic activation under visible light drove aggressive tussling. Using a further modified aggression assay, GH males exhibit increased tussling and maintain territorial control, which could contribute to a mating advantage over SH males, although direct measures of reproductive success are still needed

Strengths:

Through a series of clever neurogenetic and behavioral approaches, the authors implicate specific subsets of ORNs and pC1 neurons in promoting distinct forms of aggressive behavior, particularly tussling. They have devised a refined territorial control paradigm, which appears more robust than earlier assays using a food cup (Chen et al., 2002). This new setup is relatively clutter-free and could be amenable to future automation using computer vision approaches. The updated Figure 5, which combines cell-type-specific developmental manipulation of pC1SS2 neurons with behavioral output, provides a link between developmental mechanisms and functional aggression circuits. The manuscript is generally well written, and the claims are largely supported by the data.

Weakness:

Although most concerns have been addressed, the manuscript still lacks a rigorous, objective method for quantifying lunging and tussling. Because scoring appears to have been done manually and a single lunge in a 30 fps video spans only 2-3 frames, the 0.2 s cutoff seems arbitrary, and there are no objective criteria distinguishing reciprocal lunging from tussling. Despite this, the study offers valuable insights into the neural and behavioral mechanisms of *Drosophila* aggression.

<https://doi.org/10.7554/eLife.104212.2.sa3>

Reviewer #2 (Public review):**Summary:**

Gao et al. investigated the change of aggression strategies by the social experience and its biological significance by using *Drosophila*. Two modes of inter-male aggression in *Drosophila* are known: lunging, high-frequency but weak mode, and tussling, low-frequency but more vigorous mode. Previous studies have mainly focused on the lunging. In this paper, the authors developed a new behavioral experiment system for observing tussling behavior and found that tussling is enhanced by group rearing, while lunging is suppressed. They then searched for neurons involved in the generation of tussling. Although olfactory receptors named Or67d and Or65a have previously been reported to function in the control of lunging, the authors found that these neurons do not function in the execution of tussling and another olfactory receptor, Or47b, is required for tussling, as shown by the inhibition of neuronal activity and the gene knockdown experiments. Further optogenetic experiments identified a small number of central neurons pC1[SS2] that induce the tussling specifically. These neurons express doublesex (*dsx*), a sex-determination factor, and knockdown of *dsx* strongly suppresses the induction of tussling. In order to further explore the ecological significance of the aggression mode change in group-rearing, a new behavioral experiment was performed to examine the territorial control and the mating competition. And finally, the authors found that differences in the social experience (group vs. solitary rearing) and the associated change in aggression strategy are important in these biologically significant competitions. These results add a new perspective to the study of aggression behavior in *Drosophila*. Furthermore, this study proposes an interesting general model in which the social experience modified behavioral changes play a role in reproductive success.

Strengths:

A behavioral experiment system that allows stable observation of tussling, which could not be easily analyzed due to its low-frequency, would be very useful. The experimental setup itself is relatively simple, just the addition of a female to the platform, so it should be applicable to future research. The finding about the relationship between the social

experience and the aggression mode change is quite novel. Although the intensity of aggression changes with the social experience was already reported in several papers (Liu et al., 2011 etc), the fact that the behavioral mode itself changes significantly has rarely been addressed, and is extremely interesting. The identification of sensory and central neurons required for the tussling makes appropriate use of the genetic tools and the results are clear. A major strength of this study in neurobiology is the finding that another group of neurons (Or47b-expressing olfactory neurons and pC1[SS2] neurons), distinct from the group of neurons previously thought to be involved in low-intensity aggression (i.e. lunging), function in the tussling behavior. Furthermore, the results showing that the regulation of aggression by pC1[SS2] neurons is based on the function of the *dsx* gene will bring a new perspective to the field. Further investigation of the detailed circuit analysis is expected to elucidate the neural substrate of the conflict between the two aggression modes. The experimental systems examining the territory control and the reproductive competition in Fig. 6 are novel and have advantages in exploring their biological significance. It is important to note that in addition to showing the effects of age and social experience on territorial and mating behaviors, the authors experimentally demonstrated that altered fighting strategy has effects with respect to these behaviors.

<https://doi.org/10.7554/eLife.104212.2.sa2>

Reviewer #3 (Public review):

In this revised manuscript, Gao et al. presented a series of well-controlled behavioral data showing that tussling, a form of high-intensity fighting among male fruit flies (*Drosophila melanogaster*) is enhanced specifically among socially experienced and relatively old males. Moreover, results of behavioral assays led authors to suggest that increased tussling among socially experienced males may increase mating success. They also concluded that tussling is controlled by a class of olfactory sensory neurons and sexually dimorphic central neurons that are distinct from pathways known to control lunges, a common male-type attack behavior.

A major strength of this work is that it is the first attempt to characterize behavioral function and neural circuit associated with *Drosophila* tussling. Many animal species use both low-intensity and high-intensity tactics to resolve conflicts. High-intensity tactics are mostly reserved for escalated fights, which are relatively rare. Because of this, tussling in the flies, like high-intensity fights in other animal species, have not been systematically investigated. Previous studies on fly aggressive behavior have often used socially isolated, relatively young flies within a short observation duration. Their discovery that 1) older (14-days old) flies tend to tussle more often than younger (2 to 7-days-old) flies, 2) group-reared flies tend to tussle more often than socially isolated flies, and 3) flies tend to tussle at later stage (mostly ~15 minutes after the onset of fighting), are the result of their creativity to look outside of conventional experimental settings. These new findings are key for quantitatively characterizing this interesting yet under-studied behavior.

Newly presented data have made several conclusions convincing. Detailed descriptions of methods to quantify behaviors help understand the basis of their claims by improving transparency. However, I remain concerned about authors' persistent attempt to link the high intensity aggression to reproductive success. The authors' effort to "tone down" the link between the two phenomena remains insufficient. There are purely correlational. I reiterate this issue because the overall value of the manuscript would not change with or without this claim.

<https://doi.org/10.7554/eLife.104212.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

This work addresses an important question in the field of Drosophila aggression and mating- prior social isolation is known to increase aggression in males by increased lunging, which is suppressed by group housing (GH). However, it is also known that single-housed (SH) males, despite their higher attempts to court females, are less successful. Here, Gao et al., developed a modified aggression assay, to address this issue by recording aggression in Drosophila males for 2 hours, over a virgin female which is immobilized by burying its head in the food. They found that while SH males frequently lunge in this assay, GH males switch to higher intensity but very low-frequency tussling. Constitutive neuronal silencing and activation experiments implicate cVA sensing Or67d neurons promoting high-frequency lunging, similar to earlier studies, whereas Or47b neurons promote low-frequency but higher intensity tussling. Using optogenetic activation they found that three pairs of pC1 neurons- pC1SS2 increase tussling. While P1a neurons, previously implicated in promoting aggression and courtship, did not increase tussling in optogenetic activation (in the dark), they could promote aggressive tussling in thermogenetic activation carried out in the presence of visible light. It was further suggested, using a further modified aggression assay that GH males use increased tussling and are able to maintain territorial control, providing them mating advantage over SI males and this may partially overcome the effect of aging in GH males.

Strengths

Using a series of clever neurogenetic and behavioral approaches, subsets of ORNs and pC1 neurons were implicated in promoting tussling behaviors. The authors devised a new paradigm to assay for territory control which appears better than earlier paradigms that used a food cup (Chen et al, 2002), as this new assay is relatively clutter-free, and can be eventually automated using computer vision approaches. The manuscript is generally well-written, and the claims made are largely supported by the data.

Thank you for your precise summary of our study, and being very positive on the novelty and significance of the study.

Weaknesses

I have a few concerns regarding some of the evidence presented and claims made as well as a description of the methodology, which needs to be clarified and extended further.

(1) Typical paradigms for assaying aggression in Drosophila males last for 20-30 minutes in the presence of nutritious food/yeast paste/females or all of these (Chen et al. 2002, Nilsen et al., 2004, Dierick et al. 2007, Dankert et al., 2009, Certel & Kravitz 2012). The paradigm described in Figure 1 A, while important and more amenable for video recording and computational analysis, seems a modification of the assay from Kravitz lab (Chen et al., 2002), which involved using a female over which males fight on a food cup. The modifications include a flat surface with a central food patch and a female with its head buried in the food, (fixed female) and much longer adaptation and recording times respectively (30 minutes, 2 hours), so in that sense, this is not a 'new' paradigm but a modification of an existing paradigm and its description as new should be

appropriately toned down. It would also be important to cite these earlier studies appropriately while describing the assay.

We now toned down the description of the paradigm and cited more related references.

(2) Lunging is described as a 'low intensity' aggression (line 111 and associated text), however, it is considered a mid to high-intensity aggressive behavior, as compared to other lower-intensity behaviors such as wing flicks, chase, and fencing. Lunging therefore is lower in intensity 'relative' to higher intensity tussling but not in absolute terms and it should be mentioned clearly.

We have modified the description as suggested.

(3) It is often difficult to distinguish faithfully between boxing and tussling and therefore, these behaviors are often clubbed together as box, tussle by Nielsen et al., 2004 in their Markov chain analysis as well as a more detailed recent study of male aggression (Simon & Heberlein, 2020). Therefore, authors can either reconsider the description of behavior as 'box, tussle' or consider providing a video representation/computational classifier to distinguish between box and tussle behaviors.

Indeed, we could not faithfully distinguish boxing and tussling. To address this concern, we now made textual changes in the result section we occasionally observed the high-intensity boxing and tussling behavior in male flies, which are difficult to distinguish and hereafter simply referred to as tussling.

We also added this information in the Materials and Methods section Tussling is often mixed with boxing, in which both flies rear up and strike the opponent with forelegs. Since boxing is often transient and difficult to distinguish from tussling, we referred to the mixed boxing and tussling behavior simply as tussling.

(4) Simon & Heberlein, 2020 showed that increased boxing & tussling precede the formation of a dominance hierarchy in males, and lunges are used subsequently to maintain this dominant status. This study should be cited and discussed appropriately while introducing the paradigm.

We now cited this important study in both the Introduction and Discussion sections.

(5) It would be helpful to provide more methodological details about the assay, for instance, a video can be helpful showing how the males are introduced in the assay chamber, are they simply dropped to the floor when the film is removed after 30 minutes (Figures 1-2)?

We now provided more detailed description about behavioral assays and how we analyze them. For example All testers were loaded by cold anesthesia. After a 30-minute adaptation, the film was gently removed to allow the two males to fall into the behavioral chamber, and the aggressive behavior was recorded for 2 hours.

(6) The strain of Canton-S (CS) flies used should be mentioned as different strains of CS can have varying levels of aggression, for instance, CS from Martin Heisenberg lab shows very high levels of aggressive lunges. Are the CS lines used in this study isogenized? Are various genetic lines outcrossed into this CS background? In the methods, it is not clear how the white gene levels were controlled for various aggression experiments as it is known to affect aggression (Hoyer et al. 2008).

We used the wtcs flies from Baker lab in Janelia Research Campus, and are not sure where they are originated. We appreciate your concern on the use of wild-type strains as they may show different fighting levels, but this study mainly used wild-type strains to compare behavioral differences between SH and GH males. All flies tested in this study are in w+ background, based on w+ balancers flies but are not backcrossed. We have listed detailed genotypes of all tested flies in Table S1 in the revised manuscript.

(7) How important it is to use a fixed female for the assay to induce tussling? Do these females remain active throughout the assay period of 2.5 hours? Is it possible to use decapitated virgin females for the assay? How will that affect male behaviors?

We used a fixed female to restrict it in the center of food. These females remain active throughout the assay as their legs and abdomens can still move. Such design intends to combine the attractive effects from both female and food. One can also use decapitated females, but in this case, males can push the decapitated female into anywhere in the behavioral chamber. The logic to use fixed females has now been added in the Materials and Methods section of the revised manuscript.

(8) Raster plots in Figure 2 suggest a complete lack of tussling in SH males in the first 60 minutes of the encounter, which is surprising given the longer duration of the assay as compared to earlier studies (Nielsen et al. 2004, Simon & Heberlein, 2020 and others), which are able to pick up tussling in a shorter duration of recording time. Also, the duration for tussling is much longer in this study as compared to shorter tussles shown by earlier studies. Is this due to differences in the paradigm used, strain of flies, or some other factor? While the bar plots in Figure 2D show some tussling in SH males, maybe an analysis of raster plots of various videos can be provided in the main text and included as a supplementary figure to address this.

Indeed, tussling is very low in SH males in our paradigm, which may be due to different genetic backgrounds and behavioral assays. Since tussling behavior is a rare fighting form, it is not surprising to see variation between studies from different labs. Nevertheless, this study compared tussling behaviors in SH and GH males, and our finding that GH males show much more tussling behaviors is convincing. The longer duration of tussling in our paradigm may also be due to the modified behavioral paradigm, which also supports that tussling is a high-level fighting form.

(9) Neuronal activation experiments suggesting the involvement of pC1SS2 neurons are quite interesting. Further, the role of P1a neurons was demonstrated to be involved in increasing tussling in thermogenetic activation in the presence of light (Figure 4, Supplement 1), which is quite important as the role of vision in optogenetic activation experiments, which required to be carried out in dark, is often not mentioned. However, in the discussion (lines 309-310) it is mentioned that PC1SS2 neurons are 'necessary and sufficient' for inducing tussling. Given that P1a neurons were shown to be involved in promoting tussling, this statement should be toned down.

Thank you for this important comment. We now toned down the statement on pC1SS2 function.

(10) Are Or47b neurons connected to pC1SS2 or P1a neurons?

We conducted pathway analysis in the FlyWire electron microscopy database to investigate the connection between Or47b neurons and pC1 neurons. The results indicate that at least three levels of interneurons are required to establish a connection from Or47b neurons to

pC1 neurons. Although the FlyWire database currently only contains neuronal data from female brains, they provide a reference for circuit connect in males.

(11) The paradigm for territory control is quite interesting and subsequent mating advantage experiments are an important addition to the eventual outcome of the aggressive strategy deployed by the males as per their prior housing conditions. It would be important to comment on the 'fitness outcome' of these encounters. For instance, is there any fitness advantage of using tussling by GH males as compared to lunging by SH males? The authors may consider analyzing the number of eggs laid and eclosed progenies from these encounters to address this.

Thank you for this suggestion. We agree with you and other reviewers that increased tussling behaviors correlate with better mating competition, but it is difficult for us to make a direct link between them. Thus, in the revised manuscript, we prefer to tone down this statement but not expanding on this part.

Reviewer #2 (Public review):

Summary

Gao et al. investigated the change of aggression strategies by the social experience and its biological significance by using Drosophila. Two modes of inter-male aggression in Drosophila are known lunging, high-frequency but weak mode, and tussling, low-frequency but more vigorous mode. Previous studies have mainly focused on the lunging. In this paper, the authors developed a new behavioral experiment system for observing tussling behavior and found that tussling is enhanced by group rearing while lunging is suppressed. They then searched for neurons involved in the generation of tussling. Although olfactory receptors named Or67d and Or65a have previously been reported to function in the control of lunging, the authors found that these neurons do not function in the execution of tussling, and another olfactory receptor, Or47b, is required for tussling, as shown by the inhibition of neuronal activity and the gene knockdown experiments. Further optogenetic experiments identified a small number of central neurons pC1[SS2] that induce the tussling specifically. In order to further explore the ecological significance of the aggression mode change in group rearing, a new behavioral experiment was performed to examine territorial control and mating competition. Finally, the authors found that differences in the social experience (group vs. solitary rearing) are important in these biologically significant competitions. These results add a new perspective to the study of aggressive behavior in Drosophila. Furthermore, this study proposes an interesting general model in which the social experience-modified behavioral changes play a role in reproductive success.

Strengths

A behavioral experiment system that allows stable observation of tussling, which could not be easily analyzed due to its low frequency, would be very useful. The experimental setup itself is relatively simple, just the addition of a female to the platform, so it should be applicable to future research. The finding about the relationship between the social experience and the aggression mode change is quite novel. Although the intensity of aggression changes with the social experience was already reported in several papers (Liu et al., 2011, etc), the fact that the behavioral mode itself changes significantly has rarely been addressed and is extremely interesting. The identification of sensory and central neurons required for the tussling makes appropriate use of the genetic tools and the results are clear. A major strength of the neurobiology in this study is the finding that another group of neurons (Or47b-expressing olfactory neurons and pC1[SS2] neurons), distinct from the group of neurons previously thought to be involved in low-intensity

aggression (i.e. lunging), function in the tussling behavior. Further investigation of the detailed circuit analysis is expected to elucidate the neural substrate of the conflict between the two aggression modes.

Thank you for the acknowledgment of the novelty and significance of the study, and your suggestions for improving the manuscript.

Weaknesses

The experimental systems examining the territory control and the reproductive competition in Figure 5 are novel and have advantages in exploring their biological significance. However, at this stage, the authors' claim is weak since they only show the effects of age and social experience on territorial and mating behaviors, but do not experimentally demonstrate the influence of aggression mode change itself. In the Abstract, the authors state that these findings reveal how social experience shapes fighting strategies to optimize reproductive success. This is the most important perspective of the present study, and it would be necessary to show directly that the change of aggression mode by social experience contributes to reproductive success.

We agree that our data did not directly show that it is the change of aggression mode that results in territory and reproductive advantages in GH males. To address the concern, we have toned down the statement throughout the manuscript. For example, we made textual changes in the abstract as following

Moreover, shifting from lunging to tussling in socially enriched males is accompanied with better territory control and mating success, mitigating the disadvantages associated with aging. Our findings identify distinct sensory and central neurons for two fighting forms and suggest how social experience shapes fighting strategies to optimize reproductive success.

In addition, a detailed description of the tussling is lacking. For example, the authors state that the tussling is less frequent but more vigorous than lunging, but while experimental data are presented on the frequency, the intensity seems to be subjective. The intensity is certainly clear from the supplementary video, but it would be necessary to evaluate the intensity itself using some index. Another problem is that there is no clear explanation of how to determine the tussling. A detailed method is required for the reproducibility of the experiment.

Thank you for this important suggestion. We now analyzed duration of tussling and lunging, and found that a lunging event is often very short (less than 0.2s), while a tussling event may last from seconds to minutes. This new data is added as Figure 2G. In addition, we also provided more detailed methods regarding to tussling behavior

Reviewer #3 (Public review):

*In this manuscript, Gao et al. presented a series of intriguing data that collectively suggest that tussling, a form of high-intensity fighting among male fruit flies (*Drosophila melanogaster*) has a unique function and is controlled by a dedicated neural circuit. Based on the results of behavioral assays, they argue that increased tussling among socially experienced males promotes access to resources. They also concluded that tussling is controlled by a class of olfactory sensory neurons and sexually dimorphic central neurons that are distinct from pathways known to control lunges, a common male-type attack behavior.*

*A major strength of this work is that it is the first attempt to characterize the behavioral function and neural circuit associated with *Drosophila* tussling. Many animal species use both low-intensity and high-intensity tactics to resolve conflicts. High-intensity tactics are mostly reserved for escalated fights, which are relatively rare. Because of this, tussling in the flies, like high-intensity fights in other animal species, has not been systematically investigated. Previous studies on fly aggressive behavior have often used socially isolated, relatively young flies within a short observation duration. Their discovery that 1) older (14-days-old) flies tend to tussle more often than younger (2-days-old) flies, 2) group-reared flies tend to tussle more often than socially isolated flies, and 3) flies tend to tussle at a later stage (mostly ~15 minutes after the onset of fighting), are the result of their creativity to look outside of conventional experimental settings. These new findings are keys for quantitatively characterizing this interesting yet under-studied behavior.*

Precisely because their initial approach was creative, it is regrettable that the authors missed the opportunity to effectively integrate preceding studies in their rationale or conclusions, which sometimes led to premature claims. Also, while each experiment contains an intriguing finding, these are poorly related to each other. This obscures the central conclusion of this work. The perceived weaknesses are discussed in detail below.

Thank you for the precise summary of the key findings and novelty of the study, and your insightful suggestions.

Most importantly, the authors' definition of "tussling" is unclear because they did not explain how they quantified lunges and tussling, even though the central focus of the manuscript is behavior. Supplemental movies S1 and S2 appear to include "tussling" bouts in which 2 flies lunge at each other in rapid succession, and supplemental movie S3 appears to include bouts of "holding", in which one fly holds the opponent's wings and shakes vigorously. These cases raise a concern that their behavior classification is arbitrary. Specifically, lunges and tussling should be objectively distinguished because one of their conclusions is that these two actions are controlled by separate neural circuits. It is impossible to evaluate the credibility of their behavioral data without clearly describing a criterion of each behavior.

Thank you for this very important suggestion. We now provided more detailed description of the two fighting forms in the Materials and Methods section. See below

Lunging is characterized by a male raising its forelegs and quickly striking the opponent, and each lunge typically lasts less than 0.2 seconds through detailed analysis. Tussling is characterized by both males using their forelegs and bodies to tumble over each other, and this behavior may last from seconds to minutes. Tussling is often mixed with boxing, in which both flies rear up and strike the opponent with forelegs. Since boxing is often transient and difficult to distinguish from tussling, we referred to the mixed boxing and tussling behavior simply as tussling. As we manually analyze tussling for 2 hours for each pair of males, it is possible that we may miss some tussling events, especially those quick ones.

It is also confusing that the authors completely skipped the characterization of the tussling-controlling neurons they claimed to have identified. These neurons (a subset of so-called pC1 neurons labeled by previously described split-GAL4 line pC1SS2) are central to this manuscript, but the only information the authors have provided is its gross morphology in a low-resolution image (Figure 4D, E) and a statement that "only 3 pairs of pC1SS2 neurons whose function is both necessary and sufficient for inducing tussling in males" (lines 310-311). The evidence that supports this claim isn't provided. The expression pattern of pC1SS2 neurons in males has been only briefly described in reference 46. It is possible that these neurons overlap with previously characterized dsx+

and/or fru+ neurons that are important for male aggressions (measured by lunges), such as in Koganezawa et al., Curr. Biol. 2016 and Chiu et al., Cell 2020. This adds to the concern that lunge and tussling are not as clearly separated as the authors claim.

Thank you very much for this important question. Indeed, there are many experiments that could do to better understand the function of pC1SS2 neurons, and we only provide the initial characterization of them due to the limited scope of this study. My lab has been focused on studying P1/pC1 function in both male and female flies and will continue to do so.

To partially address your concern, we made the following revisions

- (1) We provided higher-resolution images of P1a and pC1SS2 (Figure 4C-4E). While their cell bodies are very close, they project to distinct brain regions, in addition to some shared ones.
- (2) By staining these neurons with GFP and co-staining with anti-FruM or anti-DsxM antibodies, we showed that P1a neurons are partially FruM-positive and partially DsxM-positive, while pC1SS2 neurons are DsxM-positive and FruM-negative (Figure 5A-5D).
- (3) As pC1SS2 neurons are DsxM-positive and FruM-negative, we also examined how DsxM regulates the development of these neurons. We found that knocking down DsxM expression in pC1SS2 neurons using RNAi significantly affected pC1 development regarding to both cell numbers (Figure 5G) and their projections (Figure 5H).
- (4) We further found that DsxM in pC1SS2 neurons is crucial for executing their tussling-promoting function, as optogenetic activation of these neurons with DsxM knockdown failed to induce tussling behavior in the initial activation period, and a much lower level of tussling in the second activation period compared to control males (Figure 5I-5K).
- (5) While it is very difficult to identify the upstream and downstream neurons of P1a and pC1SS2 neurons, we made an initial step by utilizing trans-tango and retro-Tango to visualize potential downstream and upstream neurons of P1a and pC1SS2 (Figure 4-figure supplement 2), which certainly needs future investigation.

While their characterizations of tussling behaviors in wild-type males (Figures 1 and 2) are intriguing, the remaining data have little link with each other, making it difficult to understand what their main conclusion is. Figure 3 suggests that one class of olfactory sensory neurons (OSN) that express Or47b is necessary for tussling behavior. While the authors acknowledged that Or47b-expressing OSNs promote male courtship toward females presumably by detecting cuticular compounds, they provided little discussion on how a class of OSN can promote two different types of innate behavior. No evidence of a functional or circuitry relationship between the Or47b pathway and the pC1SS2 neurons was provided. It is unclear how these two components are relevant to each other.

It has been previously found that Or47b-expressing ORNs respond to fly pheromones common to both sexes, and group-housing enhances their sensitivity. Regarding to how Or47b ORNs promotes two different types of innate behaviors, a simple explanation is that they act on multiple second-order and further downstream neurons to regulate both courtship and aggression, not mentioning that neural circuitries for courtship and aggression are partially shared. We did not include this in the discussion as we would like to focus on aggression modes, and how different ORNs (Or47b and Or67d) mediate distinct aggression modes.

Regarding to the relationship between Or47b ORNs and pC1SS2 neurons, or in general ORNs to P1/pC1, it is interesting and important to explore, but probably in a separate study. We tried to conduct pathway connection analyses from Or47b to pC1 using the FlyWire database, and found that Or47b neurons can act on pC1 neurons via three layers of interneurons. Although the FlyWire database currently only contains neuronal data from female brains,

they can provide a certain degree of reference. We hope the editor and reviewers would agree with us that identifying these intermediate neurons involved in their connection is beyond this study.

Lastly, the rationale of the experiment in Figure 5 and the interpretation of the results is confusing. The authors attributed a higher mating success rate of older, socially experienced males over younger, socially isolated males to their tendency to tussle, but tussling cannot happen when one of the two flies is not engaged. If, for instance, a socially isolated 14-day-old male does not engage in tussling as indicated in Figure 2, how can they tussle with a group-housed 14-day-old male? Because aggressive interactions in Figure 5 were not quantified, it is impossible to conclude that tussling plays a role in copulation advantage among pairs as authors argue (lines 282-288).

Indeed, we do not have direct evidence to show it is tussling that makes socially experienced males to dominate over socially isolated males. To address your concern, we have made following revisions

- (1) We toned down the statements about the relationship between fighting strategies and reproductive success throughout the manuscript. For example, in the abstract Moreover, shifting from lunging to tussling in socially enriched males is accompanied with better territory control and mating success.
- (2) Regarding to whether a SH male can engage in tussling with a GH male, we found that while two SH males rarely perform tussling, paired SH and GH males displayed similar levels of tussling like two GH males, although tussling duration from paired SH and GH males is significantly lower compared to that in two GH males (Figure 6-figure supplement 2).
- (3) To support the potential role of tussling in territory control and mating competition, we performed additional experiments to silence Or47b or pC1SS2 neurons that almost abolished tussling, and paired these males with control males. We found that males with Or47b or pC1SS2 neurons silenced cannot compete over control males, further suggesting the involvement of tussling in territory control and mating competition.

Despite these weaknesses, it is important to acknowledge the authors' courage to initiate an investigation into a less characterized, high-intensity fighting behavior. Tussling requires the simultaneous engagement of two flies. Even if there is confusion over the distinction between lunges and tussling, the authors' conclusion that socially experienced flies and socially isolated flies employ distinct fighting strategies is convincing. Questions that require more rigorous studies are 1) whether such differences are encoded by separate circuits, and 2) whether the different fighting strategies are causally responsible for gaining ethologically relevant resources among socially experienced flies. Enhanced transparency of behavioral data will help readers understand the impact of this study. Lastly, the manuscript often mentions previous works and results without citing relevant references. For readers to grasp the context of this work, it is important to provide information about methods, reagents, and other key resources.

Thank you very much for this comment and we almost totally agree.

- (1) Our results suggest the involvement of distinct sensory neurons and central neurons for lunging and tussling, but do not exclude the possibility that they may also utilize shared neurons. For example, activation of P1a neurons promotes both lunging and tussling in the presence of light.
- (2) We have now toned down the statements about the relationship between fighting strategies and reproductive success throughout the manuscript.

(3) We provided more detailed methods, genotypes of flies to improve transparency of the manuscript.

Reviewer #1 (Recommendations for the authors):

(1) Figure 1 Supplement 1 shows that increased aging has a linear and inverse relationship with the number of lunges, this is in contrast to a previous study from Dierick lab (Chowdhury, 2021), where using Divider assays they showed that aggressive lunges increased up to day 10 and subsequently decreased in 30-day old flies. Given that this study did not use 14-day-old flies, it might be useful to comment on this.

Thank you for this comment. Indeed, Chowdhury et al., suggested a decline of lunging after 10 days, which is not contradictory to our findings that lunging in 14d-old males is lower than that in 7d-old males. It is ideally to perform a time-series experiments to reveal the detailed relationship between ages and aggression (lunging or tussling) levels, but given our initial findings that 14d-old males showed stable tussling behavior, we prefer to use this time point for the rest of this study.

(2) For Figure 3, do various manipulations also affect the duration of tussling and boxing besides frequency and latency?

Thank you for this comment. We only analyzed latency and frequency, but not duration, as data analysis was performed manually rather than automatically on every fly pair for about 2 hours, which is very labor-consuming. We hope you could agree with us that the two parameters (frequency and latency) for tussling are representative for assaying this behavior.

(3) For Figure 3 A-F, the housing status of the males is not clearly mentioned either in the main text or the figure. What is the status of the tussling and lunging status when this housing condition is reversed when Or47b neurons are silenced, or the gene is knocked down? Do these manipulations overcome the effect of housing conditions similar to what is seen in NaChBac-mediated activation experiments?

Figure 3A-F used group-housed males and we have now added such information in the figure legends as well as Table S1.

We appreciate your suggestion on using different housing conditions. As silencing Or47b neurons or knocking down Or47b reduced tussling, it is reasonable to use GH males (as we did in Figure 3A-F) that performed stable tussling behavior, but not SH males that rarely tussle.

(4) The connections between Or47b neurons and pC1SS2 or P1a neurons can be addressed by available connectomic datasets or TransTango/GRASP approaches.

Thank you for this important suggestion. We used the FlyWire electron microscope database to analyze the pathway connections between these two types of neurons. The results indicated that there are at least three levels of interneurons for connecting Or47b and pC1 neurons. Although the FlyWire database currently only contains neuronal data from female brains, they can provide a certain degree of reference for males.

The lack of direct synaptic connection also suggests that it is challenging to resolve the connection between these two neuronal types using methods like trans-Tango/GRASP. To partially address this question, we utilized trans-Tango and retro-Tango techniques to visualize potential downstream and upstream neurons of P1a and pC1SS2 (Figure 4-figure

supplement 2). Future investigations are certainly needed for clarifying functional connections between Or47b/Or67d and P1a/pC1SS2 neurons.

(5) Figure 5, 'Winning index' and 'Copulation advance index' while described in Material and Methods, should be referred to in the main text.

We now described these two indices briefly in the main manuscript, and in the Discussion section with more details.

(6) Figure 6 shows comparisons for territorial control and mating outcomes where four different housing and aging conditions are organized in a hierarchical sequence. It is not clear from the data in Figure 5, how this conclusion was arrived at. A supplementary table with various outcomes with statistical analysis would help with this.

We now added a supplementary table (Table S2) with various outcomes with statistical analysis.

Minor Comments

(1) Line 26 says that the courtship levels in SH and GH males are not different, however, unilateral wing extension is higher in SH males as compared to GH males (Pan & Baker, 2014; Inagaki et al., 2014), also it was shown that courtship attempts are higher in *D. paulistorium* (Kim & Ehrman, 1998). It would be better to clarify this statement.

Indeed, it is found in some cases that SH males court more vigorously than GH males. We have added more references on this matter in the introduction.

(2) Figure 4, correct 'Tussing' to 'Tussling' or 'Box, Tussling' as appropriate.

Corrected.

(3) Duistermars, 2018 should be cited while discussing the role of vision in aggression (Figure 4). [A Brain Module for Scalable Control of Complex, Multi-motor Threat Displays]

We now cited this reference and added more discussion in the revised manuscript.

(4) Reviews on *Drosophila* aggression and social isolation can be cited in the introduction/discussion to incorporate recent literature e.g., Palavicino-Maggio, 2022 [The Neuromodulatory Basis of Aggression Lessons From the Humble Fruit Fly]; Yadav et al., 2024[Lessons from lonely flies Molecular and neuronal mechanisms underlying social isolation], etc.

We now cited these references in both the introduction and discussion sections.

(5) The concentration of apple juice agar should be mentioned in the methods.

We added this and other necessary information for materials in the Materials and Methods section of the study.

(6) Source of the LifeSongX software and, if available, a Github link would be helpful to include in the materials and methods section.

We now provided the source of the LifesongY software (website <https://sourceforge.net/projects/lifesongy/>), which is a Windows version of LifesongX

(Bernstein, Adam S. et al., 1992).

Reviewer #2 (Recommendations for the authors):

(1) Major comment 1

As pointed out in the public review, the weakness of this study is that the relationship between the aggression strategy and reproductive success is an inference that is not based on experimental facts; I understand that the frequency of tussling is not so high, but at least tussling-like behavior can be observed in the territory control experiment shown in Video 3. Wouldn't it be possible to re-analyse data and examine the correlation between aggressive behavior and territory control? Even if the analysis of tussling itself in this setup is difficult, for example, additional experiments using Or47b knock-out fly or pC1[SS2]-inactivated fly could provide stronger support.

Indeed, we can only make a correlation between the type of aggressive behavior and territory control. We now toned down this statement throughout the manuscript. For example, in the abstract, we changed our conclusions as following

Moreover, shifting from lunging to tussling in socially enriched males is accompanied with better territory control and mating success. Our findings identify distinct sensory and central neurons for two fighting forms and suggest how social experience shapes fighting strategies to optimize reproductive success.

To further address the concern, we now performed additional experiments to silence Or47b or pC1SS2 neurons that almost abolished tussling, and paired these males with control males. We found that males with Or47b or pC1SS2 neurons silenced cannot compete over control males (Figure 6-figure supplement 3), further suggesting the involvement of tussling in territory control and mating competition.

In relation to the above, some of the text in the Abstract should be changed. Line 28 These findings "reveal" how social experience shapes fighting strategies to optimise reproductive success.

"suggest" is more accurate at this stage.

Changed as suggested.

(2) Major comment 2

The tussling is the central subject of this paper. However, neither the main text nor Materials and Methods section provides a clear explanation of how this aggression mode was detected. Did the authors determine this behavior manually? Or was it automatically detected by some kind of image analysis? In either case, the criteria and method for detecting the tussling should be clearly described.

The behavioral data analysis in this study was performed manually. We now provided more detailed description of the two fighting forms in the Materials and Methods section. See below

Lunging is characterized by a male raising its forelegs and quickly striking the opponent, and each lunge typically lasts less than 0.2 seconds through detailed analysis. Tussling is characterized by both males using their forelegs and bodies to tumble over each other, and this behavior may last from seconds to minutes. Tussling is often mixed with boxing, in which both flies rear up and strike the opponent with forelegs. Since boxing is often transient and difficult to distinguish from tussling, we referred to the mixed boxing and tussling

behavior simply as tussling. As we manually analyze tussling for 2 hours for each pair of males, it is possible that we may miss some tussling events, especially those quick ones.

For the experimental groups where tussling cannot be observed, the latency is regarded as 120 min, but this is a value depending on the observation time. While it is reasonable to use the latency to evaluate the behavior such as the lunging that is observed at relatively early times, care should be taken when using it to evaluate the tussling. Since similar trends to those obtained for the latency are observed for Number of tussles and % of males performing tussling, it may be better to focus on these two indices.

We initially intended to provide all three statistical metrics. However, we found that using the "% of males performing tussling" would require a significantly larger sample size for subsequent statistical analysis (using chi-square tests), greatly increasing the workload. At the same time, we believe that the trend observed with "% of males performing tussling" is consistent with the other two indices, and the percentage information can also be derived from the individual sample scatter data of the other two metrics. Therefore, we opted to use "latency" and "numbers" as the statistical metrics, despite the caveat as you mentioned.

The authors repeatedly mention that tussling is less frequent but more vigorous. The low frequency can be understood from the data in Fig. 1 and Fig. 2, but there are no measured data on the intensity. As the authors mention in line 125, each tussling event appears to be sustained for a relatively long period, as can be seen from the ethogram in Fig. 2. For example, it would be possible to evaluate the intensity by measuring the duration of the tussling event.

Thank you for your valuable suggestion. We now analyzed duration of tussling and lunging, and found that a lunging event is often very short (less than 0.2s), while a tussling event may last from seconds to minutes, further supporting their relative intensities. This new data is added as Figure 2G.

(3) Minor comments

a) Line 117 How many flies were placed in one vial for group-rearing (GH)? Were males and females grouped together? Please specify in the Materials and Methods section.

We have added this information in the Materials and Methods section. In brief, 30-40 virgin males were collected after eclosion and group-housed in each food vial.

b) Line 174 The trans-Tango is basically a postsynaptic cell labeling technique. It is unlikely that the labeling intensity changes depending on neuronal activity. Do the authors want to say in this text the high activity of Or47b-expressing neurons under GH conditions? Or are they trying to show that the expression level of the Or47b gene, which is supposedly monitored by the expression of GAL4, is increased by GH conditions? The authors should clarify which is the case.

Although the primary function of the trans-Tango technique is to label downstream neurons, the original literature indicates that the signal strength in downstream neurons depends on the use of upstream neurons evidenced by age-dependent trans-Tango signals. Therefore, the trans-Tango technique can indirectly reflect the usage of upstream neurons. Our findings that GH males showed broader Or47b trans-Tango signals than SH males can indirectly suggest that group-housing experience acts on Or47b neurons. We made textually changes to clarify this.

c) Line 178 Which fly line labels the mushroom body; R19B03-GAL4?

Yes, we now provided the detailed genotypes for all tested flies in the Table S1.

d) Line 184 It was reported in Koganezawa et al., 2016 that some dsx-expressing pC1 neurons are involved in aggressive behavior. The authors should also refer to this paper as they include tussling in the observed aggressive behavior.

Thank you for this comment, and we now cited this reference in the revised manuscript.

e) Line 339 I think you misspelled fruM RNAi.

Thank you for pointing this out. fruMi refers to microRNAi targeting fruM, and we have now clearly stated this information in the main text.

f) Line 681 Is tussling time (%) the total duration of tussling occurrences during the observation time? Or is it the percentage of individuals observed tussling during the observation time? This needs to be clarified.

It is the former one. We now clearly stated this definition in the Materials and Methods section

Reviewer #3 (Recommendations for the authors):

For authors to support their conclusion that enhanced tussling among socially experienced flies allows them to better retain resources, it is necessary to quantify aggressive behaviors (mainly tussling and lunging) in Figure 5.

We agree that we can only make a correlation between enhanced tussling behavior and mating competition. We now toned down this statement throughout the manuscript. For example, in the abstract, we changed our conclusions as following Moreover, shifting from lunging to tussling in socially enriched males is accompanied with better territory control and mating success. Our findings identify distinct sensory and central neurons for two fighting forms and suggest how social experience shapes fighting strategies to optimize reproductive success.

To further address the concern, we now performed additional experiments to silence Or47b or pC1SS2 neurons that almost abolished tussling, and paired these males with control males. We found that males with Or47b or pC1SS2 neurons silenced cannot compete over control males (Figure 6-figure supplement 3), further suggesting the involvement of tussling in territory control and mating competition.

In contrast to the authors' data in Figure 4, movies in ref 36 clearly show instances of 2 flies exchanging lunges after the optogenetic activation of P1a neurons, like the examples shown in supplementary movies S1-S3. It is a clear discrepancy that requires discussion (and raises a concern about the lack of transparency about behavioral quantification).

In our study, optogenetic activation of P1^a neurons failed to induce obvious tussling behavior, and temperature-dependent activation of P1^a neurons can only induce tussling in the presence of light. These data are different from Hoopfer et al., (2015), but are generally consistent with a new study (Sten et al., Cell, 2025), in which pC1SS2 neurons but not P1a neurons promote aggression. Such discrepancy has now been discussed in the revised manuscript.

The authors often fail to cite relevant references while discussing previous results, which compromises the scholarship of the manuscript. Examples include (but are not limited to)

(1) Line 85-86 Simon and Heberlein, J. Exp. Biol. 223 jeb232439 (2020) suggested that tussling is an important factor for flies to establish a dominance hierarchy.

Reference added.

(2) Line 142-143 Cuticular compounds such as palmitoleic acid are characterized to be the ligands of Or47b by ref #18.

Reference added.

(3) Line 185-187 pC1SS1 and pC1SS2 are first characterized by ref #46. Expression data of this paper also implies that pC1SS1 and pC1SS2 label different neurons in the male brain.

We have now added this reference at the appropriate place in the revised manuscript. In addition, we have clarified that these two drivers exhibit sexually dimorphic expression patterns in the brain.

(4) Line 196-199 Cite ref #36, which describes the behavior induced by the optogenetic activation of P1a neurons.

Reference added.

(5) Line 233-235 The authors' observation that control males do not form a clear dominance directly contradicts previous observations by others (Nilsen et al., PNAS 10112342 (2002); Yurkovic et al., PNAS 10317519 (2006); also see Trannoy et al., PNAS 1134818 (2016) and Simon and Heberlein above). The authors must at least discuss why their results are different.

There is a misunderstanding here. We clearly state that there is a 'winner takes all' phenomenon. However, for wild-type males of the same age and housing condition, we calculated the winning index as (num. of wins by unmarked males – num. of wins by marked males)/10 encounters * 100%, which is roughly zero due to the randomness of marking.

(6) Line 251-254 The authors' observation that aged males are less competitive than younger males contradicts the conclusion in ref #18. Discussion is required.

We have now added a discussion on this matter. In brief, Lin et al., showed that 7d-old males are more competitive than 2d-old males, which is probably due to different levels of sexual maturity of males, but not a matter of age like our study that used up to 21d-old males.

(7) Line 274-275 It is unclear which "previous studies" "have found that social isolation generally enhances aggression but decreases mating competition in animal models". Cite relevant references.

Reference added.

(8) Line 309-310 The evidence supporting the statement that "there are only three pairs of pC1SS2 neurons". If there is a reference, cite it. If it is based on the authors' observation, data is required.

We have now provided additional data on the number of pC1SS2 neurons in Figure 5G of the revised manuscript.

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